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Industrialized Suicidal Tendencies:

An Individual Level Study of Suicide Risk in Stockholm, 1878-1925



Abstract

While economic history has a tradition of studying industrializations effects on inequality in terms of income, living standards and mortality, it has dedicated less attention to the subject of mental wellbeing. To contribute to the historical study of wellbeing, inequality and industrialization, this study uses suicide risk as a proxy for mental wellbeing. By employing a Cox proportional hazard model on a database consisting of more than 1 million individuals and 3159 suicides, the thesis was able to investigate suicide risk differentiated by social class in Stockholm, 1878-1925. The research design was guided by both classic and modern sociological theories on suicide, using key concepts such as social disintegration, disruption and dislocation to isolate variables of interest. The decision to synthesize economic historical literature with sociological theory was motivated by the latter's established history with the subject.

Results showed that even though suicide rates were differentiated by social class, rising from lowest to second highest class, suicide risk due to the theoretically identified indicators differed marginally. Chronic social dislocation through residential mobility was found to be the strongest predictor of suicide risk among all classes, a finding which was consistent and robust to competing risk. However as there likely are several confounding factors behind the finding and given the differences in rates and other covariates, the thesis concluded that the strong association between residential mobility and suicide risk should thus merit further investigation as to further ascertain inequalities in mental wellbeing.

Keywords

Suicide, suicide risk, industrialization, urbanization, social class, inequality, Stockholm, anomie, wellbeing, Durkheim

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1. Introduction

In 1884, Anna Karlsdotter moved into Stockholm from Rättvik, Dalarna. Like many migrants who arrived in Stockholm, she ended up in the predominantly working-class district of Södermalm. She was pregnant at the time and gave birth just a week after arriving. The child passed shortly after. In 1886, she moved in with Nils-Erik, a brewery worker who had just moved to Stockholm from the southern part of Sweden. During the same year, they received a child together, and the year after, they got married. Her second child did not live to see the age of two. In the coming years, Anna started working at a sugar factory while Nils-Erik also took employment in the industrial sector. In 1903, Nils-Erik and Anna moved to separate addresses, albeit still seemingly being married. In 1908, Anna took her own life. During her 20 years of living in Stockholm, Anna moved 20 times.

Anders was born in Stockholm in 1846. During his thirties, he was enlisted in the military as a lieutenant, living in the central parts of Stockholm. In 1885, after being promoted to captain, he moved in with his wife, Hanna, to Östermalm. They employed two maids at a time during the period they lived in Östermalm. The maids never stayed for long, usually working for a year or two, before the family employed a new maid. Anders and Hanna had three children together, born in 1886, 88 and 93. In 1892, Anders was promoted to major, a title he maintained until 1903, when he was taken out of service. In 1908, Anders and Hanna moved to Vasastaden, seemingly living by themselves for a time. They did not live there for long, and one year after, they moved to Kungsholmen, where the two youngest sons joined them. After leaving Östermalm, there were never any maids who lived with them. In 1915, Anders took his own life.[1]

These are but two lives drawn from the more than one million individuals registered in Stockholm's populations register between 1878 and 1926. The backdrop of both of these lives is Stockholm, currently undergoing metamorphosis, transforming from a stagnant economy into a modern, industrial city. Towards the end of the 19th century, the city was expanding rapidly, in size, population and industrial activity. Between the years of 1870 and 1930, the city would more often than not see population increases of above three percent yearly.[2] People

[1] Names are anonymized.

[2] Joseph Molitoris & Martin Dribe, "Industrialization and inequality revisited: mortality differentials and vulnerability to economic stress in Stockholm, 1878–1926". *European Review of Economic History*, Vol. 20, No. 2 (2016), p. 178.

were arriving from the entirety of Sweden to work in the capital. Neither administrative nor sanitary instances were able to keep up with the growing population. Employment in the industrial sector in turn increased by more than five percent per year from 1870 to 1910.[3]

Industrialization has presented researchers with a contradiction. Although living conditions and absolute income have been shown to increase for most people, mental wellbeing seemingly deteriorated. This fact was interpreted by moralists of time as a sign of the city's effect on the modern mind. The new urban reality produced by the industries was believed by clergymen and moral philosophers to be a breeding ground for vice and self-destructive behaviour. The secular and progressive city would disintegrate social ties whereas traditional rural life presented a safe haven.[4] The idea was exacerbated by Emile Durkheim's seminal work on suicide along with other commentaries from the time's experts. Today, while we know that the black-and-white picture painted by 19th century moralists is not the entire truth, the fact still remains that suicide did increase.[5] As moral degeneration was assumed to be the same for everyone in the cities, the question of *who* bore the brunt of industrializations social and mental burden in the city has remained unattended.

The following study asks what it was about life in a metropolitan area during its time of industrialization that affected the mental wellbeing of its population and if those effects could be felt differently, depending on which social strata you belonged to. To do so, it uses Stockholm as a case of an industrializing city. The city underwent industrialization during the late 19th century and was at the time comparable to other urban centres in the Western world.[6] Another reason for investigating Stockholm is the availability of its vast archival sources. The foundation of the current study is *Rotemansarkivet,* the population census during Stockholm's peak industrialization, which contains over 6.3 million posts on the moving patterns of the

[3] Uno Gustafson, "Industrialismens storstad: Studier rörande Stockholms sociala, ekonomiska och demografiska struktur 1860–1910". Monografier / utgivna av Stockholms kommunalförvaltning, 37. (Diss., Stockholm University, 1976), p. 70. ISSN 0346-6035

[4] Thinkers of the time pointed toward to statistics showing that urban industrial areas displayed more incidences of suicide than rural to claim that modernity was the root of the self-destructive epidemic. While acknowledged, most would ignore the caveat that urban suicide was more likely to be reported. Howard I. Kushner, "Suicide, Gender, and the Fear of Modernity in Nineteenth-Century Medical and Social Thought", *Journal of Social History*, Vol. 26, No. 3, (1993), pp. 462-463, https://doi.org/10.1353/jsh/26.3.461.

[5] Where observable, suicide rates rose consistently in most European countries during the 19th century up until the World War I Steven Stack, "The Effect of Modernization on Suicide in Finland: 1800-1984." *Sociological Perspectives*, Vol. 36, No. 2 (1993): p. 142. https://doi.org/10.2307/1389426; Kyla Thomas, David Gunnell, "Suicide in England and Wales 1861–2007: a time-trends analysis", *International Journal of Epidemiology*, Vol. 39, No. 6 (2010): p. 1467, https://doi.org/10.1093/ije/dyq094

[6] Molitoris & Dribe, "Industrialization and inequality revisited", pp. 177-178.

city's citizens.[7] By making use of Stockholm's thorough and well-preserved population register, it is possible to follow individual lives, like Anna and Anders, in order to observe how a life would have changed during this very transformative period. The highly detailed data allows us to track how their household constellations would change, how often they moved and ultimately, whether these factors put one at an increased risk of suicide.

1.1. Research aim, purpose and questions

The purpose of this essay is to examine how industrial transformations and its associated consequences affected the mental well-being of an urban population. Whereas economic history has a long tradition of examining industrializations effect on living standards and interactions with material inequalities, it has shied away from the question of mental wellbeing. Historical material on mental health and wellbeing is scarce; hence it may be for good reason that it has not been well studied. A phenomenon which has been better registered, and is closely related with mental health, is suicide. In order to access an indicator of mental wellbeing, the current study will use suicide risk as a proxy for mental wellbeing.

Suicide, and in extension, suicide risk, are however not perfect proxies for states of mental wellbeing. Past studies have shown weaker significance between suicide risk and prevalence of depression. Usually, it will display a more complex relationship with age, gender and various socioeconomic factors moderating the effects.[8] Nevertheless, this same fact I would claim is precisely what makes suicide an interesting aspect to examine in terms of social and economic conditions. If the moderating effects are found within these factors, then a study of suicide should reveal not only mental wellbeing, but furthermore *reactions* to actively changing social and economic conditions. Imperfections notwithstanding, it remains one of the few proxies available when investigating mental wellbeing during this period and should furthermore be considered an extreme reaction. While this might entail limited data, it does indicate what sort of activity that people reacted to.[9] That said, participation in industrialization did not happen on equal terms and in investigating suicide during the time, the aim is to show

[7] SwedPop. (2026) Version 4 of Harmonised Data in SwedPop. (Version 4). Original database: Rotemannen database, SE/SSA/0032/03, Stockholms Stadsarkiv. Demographic Data Base (DDB), Umeå University. https://doi.org/10.48524/SWEDPOP-V4-0

[8] Socioeconomic factors in the following thesis concerns varying interactions with social class as measure. Marco H.D. van Leeuwen & Ineke Maas, *HISCLASS: A Historical International Social Class Scheme*, (Leuven University Press, 2011), pp. 16-27.

[9] Information on the mental wellbeing of the population was barely collected during this time. Statistics on suicide of course present their own issues of reliability, which will be accounted for in coming chapters, but to their credit, have historically been systematically collected in at least Sweden and Finland. Stack, "The Effect of Modernization on Suicide in Finland: 1800-1984.", p. 140.

how this transition was received. To do so, it will investigate whether socioeconomic, macroeconomic and demographic changes can explain variations in suicide risk across different social classes. The research questions to be asked are thus:

- *Which factors associated with urbanization drove suicide risk during industrializing Stockholm, 1878 – 1925?*
- *How did suicide risk vary by social class during industrializing Stockholm, 1878 - 1925?*

While the period is informed on the hand by the availability of the data, the material arose due to the state of Stockholm. People from all over Sweden were arriving in Stockholm, with population growth exceeding the pace of which new housing could be built.[10] A way in which the city administration handled this was by close monitoring of the population through a new administrative office called *Rotemannen*. The product of his work resulted in highly detailed documents of the citizens of Stockholm during the period of 1878 to 1926.[11] Thus the data availability aligns with Stockholm's peak urbanization. By utilizing this individual-level data, this study will analyse how urbanization might have impacted groups differently in regard to suicide.

The study will only investigate completed suicides since the data does not provide information on attempted suicide. Reasonably attempted suicides could be indicative of similar mechanisms and should preferably not be set aside.[12] The most egregious limitation however in not being able to study attempted suicide, is the loss of data on women. Suicide has consistently been shown to be more common among men. Regarding attempts on the other hand, women are usually overrepresented.[13] It is an acknowledged problem in research on suicide, and unfortunately historical documentation on attempts is presumably worse than completed suicides.

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[10] Joseph Molitoris, "Life and Death in the City: Demography and Living Standards during Stockholm's Industrialization", Lund Studies in Economic History, Vol. 73. Diss., (Lund University 2015), pp. 19-20.

[11] Data for 1926 is incomplete due to the Roteman office being shut down. As outlined by the research question, the period relevant for the study is rather 1878 – 1925. Geschwind and Fogelvik, The Stockholm Historical Database, pp. 207-210.

[12] Joe Chen, Yun Jeong Choi, Kohta Mori, Yasuyuki Sawada & Saki Sugano, "Socio-economic studies on suicide: A survey". *Journal of Economic Surveys*, Vol. 26, No. 2 (2012), p. 282. https://doi.org/10.1111/j.1467-6419.2010.00645.x

[13] While modern statistics point to this, historical data on attempts are scarce. Kushner however claims that the same trend should apply historically. Steven Stack, "Suicide: A 15-Year Review of the Sociological Literature Part I: Cultural and Economic Factors", *Suicide and Life-Threatening Behavior*, Vol. 30, No 2 (2000), p. 146; Howard I. Kushner, "Women and Suicide in Historical Perspective." *Signs*, Vol. 10, No. 3 (1985): pp. 542-543.

2. Literature review & theory

The origins of modern suicidology stem from a specific historical context, the process of industrialization and subsequent urbanization in Europe. In sociology, these processes, along secularization, are often conceptualized as modernization.[14] The succeeding sociological research that spawned from this has spent much time on verifying and refuting the theories which arose during Europe's modernization relevance on contemporary suicide rates. At the same time, sociology has been particularly light in dealing with the historical context of said theory.[15] While sociology has garnered valuable insight into what social aspects drive suicide, the historical case studies on suicide are but a few when the theoretical foundations rely so heavily on the context it was born in. Economic history has on the other hand taken great interest in the industrialization and urbanization and their welfare consequences, such as inequality and class differences. Even if the study of mortality has been a way of conducting research on historical inequality, suicide has not been a part of this story.[16] To make the most out of the sociological theories which put grand social change front and centre, historical cases need to be addressed. Economic history's focus on the processes of industrialization and urbanization provide an important foundation from which questions of what foster suicide risk can be asked. For this reason, I believe there to be value in uniting the sociological research on suicide with the study of historical economic processes. The following literature review makes an effort to synthesize perspectives from these disciplines.

2.1. Industrialization, Mortality and Inequality

The empirical study of inequality has been a mainstay subject of economic history since Simon Kuznet's *Economic Growth and Income Inequality*. He proposed that countries experiencing industrialization and urbanization would first experience a widening of inequality as a population started migrating towards the cities, moving from the not-so-productive agricultural

[14] Steven Stack, "Suicide: A 15-Year Review of the Sociological Literature Part II: Modernization and Social Integration Perspectives." *Suicide and Life-Threatening Behavior*, Vol. 30, No 2 (2000), pp. 163-165. https://doi.org/10.1111/j.1943-278X.2000.tb01074.x

[15] Bernice A. Pescosolido, "Bringing Durkheim into the Twenty-First Century: A Network Approach to Unresolved Issues in the Sociology of Suicide" in *Émile Durkheim: "Le Suicide" One Hundred Years Later*, David Lester (eds.), (Charles Press, 1994), pp. 270-271.

[16] Studies of mortality often distinguish between all-cause and cause-specific mortality. If taken into account, 'violent' deaths, such as suicide, are often categorized under "other" in cause-specific studies. Either way, they are seldom of interest even if implicitly included in studies on all-cause mortality. Enrique Debiasi & Martin Dribe "SES inequalities in cause-specific adult mortality: a study of the long-term trends using longitudinal individual data for Sweden (1813–2014)". *Eur J Epidemiol*, Vol. 35 (2020), pp. 1053–1054. https://doi.org/10.1007/s10654-020-00685-6

sector to the much more lucrative industrial sector. As more and more people moved into the cities, the trend would start to reverse and inequality would decrease.[17] A thesis which would later become known as Kuznets Curve.[18]

The relationship has since then proven to be more nuanced, but the question of how industrialization and urbanization affect inequality has remained central in economic history. Whether it be through redistribution of income, wealth or differentials in living conditions, material inequality and its interactions have been the highlight.[19] One such interaction is that of socioeconomic differences in mortality and life expectancy. A phenomenon which recently has received a lot of attention within economic history is the socioeconomic status gradient. It refers to the specific negative association between socioeconomic indicators and health: Individuals of lower class have a significantly lower life expectancy compared to those of higher, an association which holds regardless if measuring it by education, income or social class.[20] It has been debated whether the gradient has always existed or if it emerged more recently. Most studies point to the gradient being more modern.[21] Lifestyle behaviour and differences along with psychosocial stress has been identified has the determining factor for the emergence of the gradient. Whereas lower classes were more vulnerable overall due to less access to healthcare, nutrition and sanitary facilities, lifestyle choices as smoking, alcohol consumption and inactivity were more common among higher classes. The argument being that these lifestyle behaviours later transitioned into lower classes, at the same time as public healthcare became more common.[22]

Studies on infant mortality will usually more clearly reflect the aforementioned factors of sanitation and nutrition and have been used as a measure of living conditions for this precise

[17] Simon Kuznet, "Economic Growth and Income Inequality", *The American Economic Review*, Vol. 45, No. 1 (1955), pp. 18-19.
[18] As Lindert et al. points out, Kuznet himself never drew such a curve. Peter Lindert, Wen Hai & Shunli Yao, "Three centuries of Inequality in Britain and America", *Working Paper*, No. 97-9, University of California, Department of Economic, Davis CA, p. 8.
[19] Luciana Quaranta & Maria Stanfors, "The Gender Revolution: Marriage, Fertility and Divorce in the Industrial City" in *Urban Lives: An Industrial City and Its People During the Twentieth Century*, Martin Dribe, Therese Nilsson, and Anna Tegunimataka (eds), (Oxford Academic, 2024), pp. 170-171. https://doi.org/10.1093/oso/9780197761090.001.0001
[20] Tommy Bengtsson, Martin Dribe and Jonas Helgertz, "The Late Emergence of the Socioeconomic Gradient in Adult Mortality" in *Urban Lives: An Industrial City and Its People During the Twentieth Century*, Martin Dribe, Therese Nilsson, and Anna Tegunimataka (eds), (Oxford Academic, 2024), pp. 281-282. https://doi.org/10.1093/oso/9780197761090.001.0001
[21] Most evidence suggest the gradient emerged in the mid-/late mid-20th century. To note, however, is that Debiasi investigates entire available populations and when considering cause-specific mortality, suicide is not isolated. Debiasi & Dribe, "SES inequalities in cause-specific adult mortality", pp. 1053–1054; Tommy Bengtsson, Martin Dribe & Jonas Helgertz, "When Did the Health Gradient Emerge? Social Class and Adult Mortality in Southern Sweden, 1813–2015", Demography, Vol. 57, No. 3 (2020), pp. 953–977.
[22] Debiasi & Dribe "SES inequalities in cause-specific adult mortality", pp. 1044, 1052-1053.

reason.[23] Especially pertinent in regard to studying industrialization's impact on living standards. By interacting infant mortality with SES, it has been possible to map out how inequalities in living standards developed during the industrialization. For Stockholm, data from Rotemannen have been used to show that during this period there was a social class gradient in regard to infant and child mortality.[24] Relative inequality would persist but absolute mortality on the other hand did decrease, indicating an improvement in living standards for all. Public health initiatives, accessible sanitation and nutrition have all contributed to this fact.[25]

With all of this said, I believe that by solely focusing on the material dimensions, aspects of inequality are lost. If industrialization can be said to have had a net positive effect on living standards, this conclusion presents a great contradiction. Material wellbeing and life expectancy improved at the same time as suicide increased, seemingly indicating that mental health deteriorated.[26] Whereas industrialization entailed a great economic transformation, in many aspects it was also very much a social one as well. Household constellations underwent structural changes, women were employed in higher degrees, and the population grew mainly due to migration.[27] All factors, while of course of relevance due to their impact on material living conditions, they furthermore have in modern studies shown to be relevant predictors of suicide.[28]

Suicide from the perspective of the economic historian has consequently largely been ignored and through this, a major aspect of the development of wellbeing during industrialization has been unexplored. For good reason mayhap, since inferring overall

[23] Lindert, Hai & Yao, "Three centuries of Inequality in Britain and America", p. 5.
[24] Molitoris & Dribe "Industrialization and inequality revisited", p. 184.
[25] In regard to class inequalities persisting, Molitoris and Dribe suggest that it could be due to mainly nutrition and housing standards since inequalities persisted even within districts of the city. That is to say, overcrowding in districts such as Södermalm and Kungsholmen could not explain mortality differentials. Molitoris & Dribe "Industrialization and inequality revisited", pp. 177, 192; See also; Bo Burström & Eva Bernhardt "Social differentials in the decline of child mortality in nineteenth century Stockholm", *Eur J Public Health*, Vol. 11, No. 1 (2001), p. 32-33. doi: 10.1093/eurpub/11.1.29. PMID: 11276568.
[26] Stack, "The Effect of Modernization on Suicide in Finland: 1800-1984.", p. 142.; Thomas & Gunnell, "Suicide in England and Wales 1861–2007", p. 1467.
[27] Martin Dribe & Patrick Svensson, "Landskrona: the industrial city" in *Urban Lives: An Industrial City and Its People During the Twentieth Century*, Martin Dribe, Therese Nilsson, and Anna Tegunimataka (eds), (Oxford Academic, 2024), pp. 46-48. https://doi.org/10.1093/oso/9780197761090.001.0001; Finn Hedefalk, Patrick Svensson & Anna Tegunimataka "Migration in Times of Economic Growth and Recession" in *Urban Lives: An Industrial City and Its People During the Twentieth Century*, Martin Dribe, Therese Nilsson, and Anna Tegunimataka (eds), (Oxford Academic, 2024), pp. 115-116. https://doi.org/10.1093/oso/9780197761090.001.0001
[28] Regarding specifically the association between female labour force participation and suicide, previous results are somewhat dubious. While earlier studies have made the assumption that the increase in female suicide rates during 20th century is due to higher female labour participation, recent studies suggest that the impact is concentrated in male suicide rates. Kushner, "Women and Suicide in Historical Perspective.", p. 542; Chen et al., "Socio-economic studies on suicide", pp. 286-287.

wellbeing from patterns in suicide risk can be problematic. Even if sometimes used as a proxy for happiness or mental health, suicide has throughout time consistently varied between men and women, young and old. Thus, the issue lies in generalizing across an entire population from a phenomenon that is never equally distributed. There are furthermore methodological issues in data availability. Given the nature of suicide as a highly individual phenomenon, microdata is to be preferred, although seldom available for periods of industrialization. In spite of this, I believe that these complexities strengthen the case for investigating suicide, rather than weakening it. The moderating effect of socioeconomic factors is precisely what makes suicide a meaningful indicator of social and economic differences. As for the methodological issues, the material provided by Stockholm's population register contains vast information on individual level.

Social history on the other hand has taken interest in the study of suicide and industrialization, specifically stemming from the moral panic of the 19th century reaction to the growing urban centres.[29] A study by Olive Andersson showed that in Victorian England and Wales, suicide rates in rural areas were higher than in industrial towns, with the exception of London. From this, she reaches the conclusion that industrialization did *not* increase suicides in Victorian England and that the question is more complicated than that.[30] Nevertheless, she omits offering an alternative explanation for why suicide rates did increase. Rural areas suffered intense out-migration, and the economic conditions of these places were also at risk.[31] The point stands that industrialization seemingly did affect suicide rates. However, it needs to be acknowledged that if studying the overall trend during the time, the researcher needs to look beyond the industrial cities. In any case, whether or not the increasing suicide rates were a rural response to urban activity, urban activity did change the social playing field of the cities.[32]

To conclude; social consequences closely associated with industrialization, have time and time again been showed to associate with suicide. Even so, while economic history has studied economic effects such as growth, production, mortality, development of living

[29] Kushner, "Suicide, Gender, and the Fear of Modernity in Nineteenth-Century Medical and Social Thought", pp. 461-463.
[30] Olive Anderson, "Did Suicide Increase with Industrialization in Victorian England?", *Past & Present,* No. 86 (1980), pp. 170-172.
[31] Jason Long, "Rural-Urban Migration and Socioeconomic Mobility in Victorian Britain." *The Journal of Economic History*, Vol. 65, No. 1 (2005), pp. 2-3.
[32] Industrialization saw the emergence of the Swedish industrial working class, whose share of the population grew considerably. Along with this, household sizes increased, women entered the labour force in greater numbers, being facing increasing pressure to navigate both the reproductive and productive sphere. Gustafson, "Industrialismens Storstad", pp. 25-28; Louise A. Tilly, "Women, Women's History, and the Industrial Revolution." *Social Research*, Vol. 61, No. 1 (1994), pp. 131–132.

standards and how inequalities are revealed through these factors, it has dedicated less attention how these factors relate to mental wellbeing and in extension, suicide.[33] Naturally, the study of the former is more accessible through material conditions which are easier to measure but in this process, it is easy to lose sight of individual struggle in which inequalities materialize. In this thesis I argue that suicide should be part of this story. Suicide represents a non-material welfare outcome that is able to capture mental wellbeing, normlessness and social isolation; dimensions of industrialization which material indicators cannot reach. For these reasons, I believe the study of suicide can bring valuable insights into economic historic study of industrialization. To do this fairly however, economic history needs to take lessons from sociology, a discipline with a long history of studying suicide.

2.2. Ontology of suicide and theoretical framework

Any modern study considering social, cultural or economical perspectives on suicide will inevitably touch on Emile Durkheim's work, *Le suicide*, first published in 1897. The Durkheimian understanding of suicide entails regarding suicide as a natural consequence of so-called social facts. Suicide cannot be explained by individual conditions, such as personal loss or biological causes.[34] According to Durkheim, suicide is due to two factors governing social life, *regulation* and *integration*. Regulation is the degree to which an individual's moral life is restrained by societal forces. An overabundance of regulation would put one at risk for the fatalistic suicide and, contrary, a lack of regulation would lead to the anomic suicide. Integration is the degree to which an individual is integrated into social groups and their surrounding social life. Through integration, Durkheim theorized two additional suicides, the altruistic, being too integrated, and the egoistic, not being integrated.[35] The two concepts that have had the most staying power are *integration* and *anomie* and these are the concepts that will be operationalized in the current study.

According to Durkheim, anomie is the disruption of a social life. An individual has an internalized set of norms according to social integration and regulation. Anomic suicide in turn refers to the ultimate consequence of the state of normlessness that an individual finds themselves in when said norms are disturbed. Previous empirical iterations of measuring

[33] Quaranta & Stanfors, "The Gender Revolution: Marriage, Fertility and Divorce in the Industrial City", pp. 170-171. https://doi.org/10.1093/oso/9780197761090.001.0001
[34] Émile Durkheim, *Suicide: A study in sociology*, translation by John A. Spaulding & George Simpson (Taylor & Francis, 2005), pp. 261-263.
[35] Durkheim himself disregards importance of the fatalistic suicide for the period and only mentions it in passing. The examples at the time were too few to be of any study value; Durkheim, *Suicide*, pp. 167, 175, 212-214, 239.

anomie involve the study of economic fluctuations most commonly. Durkheim proposed a non-linear relationship with economic fluctuations, meaning that both during periods of economic depression and prosperity, suicide rates would increase. The staying power of anomie can mostly be attributed to reformulations of Durkheim's original theory, notably by Robert Merton. Anomie has since been redirected towards explaining structural mismatch between individual and its surrounding cultural institutions.[36] Today, anomie is most operationalized in the context of suicide through studies on economic fluctuations and societal scale events. On the other hand, there are several studies confirming anomic relationships of suicide, in divorce rates and economic fluctuations.[37] Around the time that the World Health Organization's (WHO) acknowledged suicide as a public health issue in 2014, there has been a tradition of epidemiological studies claiming anomie.[38] While these are mostly concerned with the empirical implications, they attest to the staying power of Durkheim's theorem.[39]

The concept of integration has on the other hand become staple in sociology and suicidology. The main argument from which Durkheim observes suicides caused by a lack of integration, concerns the case of Protestants and Catholics in Europe. For this, he presented rates across protestant populations and catholic populations to show that those consisting of mainly protestants displayed higher suicide rates than that of the catholic ones. The argument being that Catholic groups were much better integrated than respective Protestant.[40] In succeeding research, family and neighbourhood have been identified as crucial aspects for integration. Whichever social sphere where an individual finds itself enmeshed, the more extensive and denser it is, the better the integration.[41] Durkheim kept anomie, and its

[36] Merton's strain theory entails the relationship between culturally defined aspirations versus social structural distribution of means to achieve aspirations. In the uneven distribution of legitimate means, strain occur. Robert K. Merton, "Social Structure and Anomie." *American Sociological Review*, Vol. 3, No. 5 (1938), p. 672-673, 681-682. https://doi.org/10.2307/2084686; Seth Abrutyn, "Toward a General Theory of Anomie: The Social Psychology of Disintegration", *European Journal of Sociology*, Vol. 60, No. 1, (2019), pp. 114-115. doi:10.1017/S0003975619000043

[37] Steven Stack, "Suicide: A 15-Year Review of the Sociological Literature Part II", pp. 167-168.

[38] WHO have since then published yearly reports on suicide; World Health Organisation, ed. Preventing Suicide: A Global Imperative. World Health Organization, 2014, p. 2.

[39] See for example; Thibodeau L, Lachaud J. "Impact of economic fluctuations on suicide mortality in Canada (1926-2008): Testing the Durkheim, Ginsberg, and Henry and Short theories". *Death Studies*, Vol. 40, No. 5 (2016), p. 305-306. doi: 10.1080/07481187.2015.1133727, or; Fejiun Luo, Curtis S. Florence, Myriam Quispe-Agnoli, Lijing Ouyang, & Alexander E. Crosby, "Impact of Business Cycles on US Suicide Rates, 1928-2007". *American Journal of Public Health*, Vol. 101, No. 6 (2011), pp. 1139-1140.

[40] The argument being that protestant teaching preaches free inquiry is the critical difference between the two religions. This in turn promotes individual though over "collective credo", thus resulting in weaker integration. Durkheim, *Suicide*, pp. 112-113.

[41] Anna S. Mueller, Seth Abrutyn, Bernice Pescosolido, Sarah Diefendorf, "The Social Roots of Suicide: Theorizing How the External Social World Matters to Suicide and Suicide Prevention", *Front Psychol*, Vol. 12 (2021), pp. 2, 8.

progenitor, regulation, separate from integration, but succeeding research has found the concept difficult to operationalize.

Even if Durkheimian theory has been the hegemon of sociological suicidology, it has not gone un-critiqued. The most repeated and glaring of issues of Durkheim's work lies in the ecological fallacy: to explain highly individual phenomena suicide through suicide rates and macro data. If one were to lift a specific example from his reasoning, such as anomie in divorce, then the issue becomes clear. If divorce is capable of explaining suicide, it is not the husband of a still-married couple that takes his own life.[42] Problems of empirical nature are also commonly lifted, on several levels. At its core, Durkheim's theory is a historical one and the results of his study should also be considered a product of its time. As he notes himself, "The entire morality of progress and perfection is thus inseparable from a certain amount of anomy".[43] However, the data on which the theory is built on mostly concerns short-term or cross-sectional evidence.[44]

Furthermore, these methodological limitations have theoretical implications. The hypothesis provided by Durkheim was that the industrializing process would create societies where individuals would neither be regulated nor integrated. So more than being a theory on why suicide happens, it is as well a historical theory on suicide as the mechanism of interest is an historical process. As the aforementioned study by Anderson points out, the theory provided by Durkheim has been in active use since its inception. Theories which are based upon the results of a highly criticized method and data. She suggests that uncritically applying concepts of anomie and integration to the historical context in which they were produced would be to play in hand the ideological moral movement of the time. Thus, historians would do well to exercise caution when dealing with the old sociological theories as to not assert assumptions of modernity being associated with degeneracy and deviance.[45] Similar critique has been levelled at sociological studies considering contemporary periods of study, but from the perspective that the world has moved on from these concepts.[46]

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[42] Seth Abrutyn & Anna S. Mueller, "The Socioemotional Foundations of Suicide: A Microsociological View of Durkheim's Suicide", *Sociological Theory*, Vol. 32, No. 4 (2014), p. 330. https://doi.org/10.1177/0735275114558633

[43] Durkheim, *Suicide*, p. 331.

[44] Stack, "The Effect of Modernization on Suicide in Finland: 1800-1984.", p. 139.

[45] As mentioned before however, Anderson does not provide an alternative explanation for the increasing suicide rates. On paper, one could theorize that increased suicide rates among rural populations could also be a product of anomie. Although not directly indicative of the phenomena Durkheim advocated for, the theoretical implications are similar. Anderson, "Did Suicide Increase with Industrialization in Victorian England?", p. 173.

[46] Pescosolido, "Bringing Durkheim into the Twenty-First Century", p. 265.

This all surmounts to the question of whether or not it is motivated to apply a theory that is more than a hundred years old, born out of a specific historical context.[47] Taking a look at the empirical work on economic and social explanations for suicide, evidence points to a somewhat divisive picture. On the one hand, explanations which Durkheim dismissed on account of them not relating to social facts, such as imitation or genetic explanations have been found to suitable predictors.[48] On the other hand, sociology has continued to use the concepts of integration and anomie with evidence pointing to real effects.

All things considered I believe a theoretical framework supported by Durkheim is a feasible alternative. The process that Durkheim described would increase the suicide rates, is precisely the one that Stockholm is undergoing at the period of interest. The concept of anomie and integration also lends themselves well to the research question at hand. Anomie in periods of industrialization would reasonably not fall evenly between social classes. As a lower-class citizen, less institutional ties and resources would make one vulnerable to the disruption of social norms. Durkheim postulated the opposite, that poverty would protect against suicide through stronger social restraint and community ties. The evidence has rather pointed to that factors such as unemployment, financial stress and family instability increase propensity for suicide. All factors associated mainly with poverty.[49] Integration would also differ depending on who you were through access to integrating institutions. In industrializing Stockholm, family, occupation and birthplace were all institutions and factors stratified across gender, age and class.[50] Nevertheless, the methodological problems and theoretical limitations outlined above, need to be acknowledged.

Where Durkheim saw anomie as a consequence of its progenitor, regulation, modern sociological theory treats integration and anomie as integral to one another. Seth Abrutyn proposes a reformulation of the theory of anomie, where instead of explaining individual level behaviour according to macro level forces, it can be regarded as a social psychological process. The process can be conceptualized as such: If integration is the main dimension of social behaviour, then social disintegration is the process leading to anomie. The disintegrating

[47] Pescosolido, "Bringing Durkheim into the Twenty-First Century", pp. 270–271.
[48] Imitation have been found to have a significant effect. Stack points to an example of a book published in 1991, a guide directed toward the terminally ill. It recommended a certain method which in turn increased the occurrences of that specific suicide the same year, with the book being found at a number of these scenes as well. Steven Stack, "Suicide: A 15-Year Review of the Sociological Literature Part I: Cultural and Economic Factors", *Suicide and Life-Threatening Behavior*, Vol. 30, No 2 (2000), pp. 148-149. https://doi.org/10.1111/j.1943-278X.2000.tb01074.x; Durkheim, *Suicide*, p. 80; Mueller, Abrutyn, Pescosolido & Diefendorf, "The Social Roots of Suicide", p. 3.
[49] Stack, "Suicide: A 15-Year Review of the Sociological Literature Part I", pp. 155-156.
[50] Gustafson, "Industrialismens Storstad", pp. 115-122.

process in turn is caused by social disruption. Thus, social disruption is the mechanism from which suicide risk increases. It can be acute, as in an immediate intense disruption, or chronic, as a long-lasting depriving anomie.[51] Abrutyn theory emphasizes shame in describing the socioemotional and sociopsychological aspects of anomie. Even though these are hard to measure in the data at hand, there is one key element he identifies that is of importance for current study. Abrutyn theorized three different categories of disruptions; social dissolution, -disjunction and -dislocation. Dissolution refers to the loss of social ties to other individuals or groups. It can materialize in divorce, bereavement or incarceration. Disjunction refers to the relationship between aspirations, expectations and actual achievement. Finally, dislocation refers to the disintegrative relationship between an individual and the loss of a physical space in which identity is formed.[52] Given the demographic and geographic nature of the data at hand, social dislocation is the key form of disruption that the present study is able to investigate.

Alternative theories with less of a theoretical connection to Durkheim, have sprung up as responses to the shortcomings of the social facts of sociology. The purely economic perspective favours the individual and its capacity to make rational choices. The economical principle behind the rational suicide is that when an individual lacks the means to improve their life, the rational choice is to end it. One of the more prominent and (in)famous texts in the economic tradition is Daniel Hamermesh and Neil Soss's article "An economic theory of suicide" (1974). Hamermesh and Soss propose that suicide is an effect of perceived utility. They formulate a function in which an individual's utility decreases by age and permanent income. If perceived utility were to reach zero then the individual in question would take their own life.[53] Even if commonly cited in more recent notable sociological research on suicide, it has not gone without criticism. For one, Hamermesh and Soss's function does not account for variation between gender or geography and why wealthy people still choose to commit suicide even though their supposed utility should be high.[54] Even if perceived to be an imperfect theory, there have been evidence supporting that a suicide *can* occur due to utility maximization.[55]

With this said, subscribing solely to ontological notions that all suicides can be explained by either social facts, as in the case of Durkheim, or through lifetime income, as in the case of Hamermesh and Soss, undercuts the complexity of suicide and is outright reductive. If our

[51] Abrutyn, "Toward a General Theory of Anomie", pp. 110, 127-128.
[52] Abrutyn, "Toward a General Theory of Anomie", pp. 123-127.
[53] Daniel S. Hamermesh, & Neil M. Soss. "An Economic Theory of Suicide". *Journal of Political Economy*, Vol. 82, No. 1 (1974), pp. 84–86.
[54] Gary N. Smith, "It is Time to Kill the Economic Theory of Suicide" (2019). *Pomona Economics*. 4. pp. 4-6.
[55] Chen brings the relatively recent case of suicide rates spiking in India due to new legislation on death insurance, although anecdotal. Chen et al., "Socio-economic studies on suicide", p. 271.

contemporary understanding of suicide is that it is a highly individual matter with several potential risk factors, then our study of past suicide must incorporate our modern understanding of the phenomena. Respect is of course paid to its cultural context in that we cannot redefine a suicide from another time. What we can do is leverage that we today, hopefully, know more about it and its mechanisms. For this reason, to make the most out of theory on suicide, a framework incorporating different aspects is most appropriate. To conclude, the Durkheimian concepts of anomie and social integration along with the modern reformulations will guide the research design of the study.

3. Operationalization

The theoretical perspective in the coming analysis has thus a foundation in Durkheim, but where relevant critique makes it non-applicable, hypotheses are formulated according to other theory. Acknowledging the ecological fallacy and given the fact that the present study is analysing individual suicides in a local context, and not suicide-rates, there has to be a partial departure of Durkheim's true ontology. Table 1 features the theories which inform the analysis, what their hypotheses entail and which measure will operationalize said theories. The theoretical framework operates under the two key concepts of social disintegration and social disruption. In operationalizing the theory, the main issue lies in deriving measures from the concepts of integration and anomie. The distinction found between the types of disintegration in table 1 is due to Durkheim's understanding of anomie and integration and Abrutyns. Durkheim regarded anomie as a macro level force where changes on a societal scale would have ramifications for individuals. On the other hand, Abrutyn's reformulation of anomie, synthesizes the concept with integration to focus on micro level psychosocial interactions. Thus, the analysis will use Abrutyn's concept of social dislocation in operationalization of the individual level variables of interest, birthplace and residential mobility.

Birthplace tells us if an individual was born in Stockholm, in another part of Sweden or in another country. Disintegration of social life can be assumed to be amplified for migrants arriving in Stockholm, being separated from their original families and neighbourhoods. Something which should be even more visible for foreign migrants, who also face differing culture and language.[56] The birthplace variable is thus indicative of acute social disruption.

[56] Stack, "Suicide: A 15-Year Review of the Sociological Literature Part II", p. 171.

Table 1: Operationalization

Theory	Hypothesis	Measure
Social disintegration on aggregated level increase suicide risk	Suicide risk has a non-linear relationship with economic fluctuations	Economic fluctuations
	Urbanization through rapid in-migration is socially disruptive	In-migration rate
Social disintegration on individual level increases suicide risk	Social dislocation is socially disruptive	Residential mobility
		Birthplace
Perceived Utility	Perceived utility decreases with age and lifetime income	Social class, Age group

Higher levels of residential mobility are hypothesized to associate with chronic social disruption through repeated social dislocation. Residential mobility has in contemporary studies been shown to associate with adverse mental and physical health effects, through substance abuse, shorter life expectancy and suicidal behaviour.[57] Most studies concerning residential mobility and suicidal behaviour have focused on adolescent groups, showing that high mobility in an early age were associated with an increased risk of suicide attempts.[58] As for the effect among adults, studies have been scarce. One study, investigating a full population in the Netherlands, found that non-movers had a higher risk of suicide than those who moved, contradicting the present study's hypothesis.[59] However, some aspects need to be distinguished between that study and the present one. Hagedoorn and Helbich's study uses a binary distinction between movers and non-movers, not accounting for the effect of continuous moving. The mechanism of interest lies in quality of neighbourhood change, operationalized as upward or downward change. Finally, the spatial and temporal context of each study is vastly different.[60] Their study, in a much higher degree, concerns selective moving, whereas moving intent or non-intent of the citizens of Stockholm in many can be harder to establish. In focusing on continuous moving, the mechanism of interest is not neighbourhood improvement but rather the intensity of social disruption.

[57] Hyewon Choi & Shigehiro Oishi, "The psychology of residential mobility: a decade of progress", *Current Opinion in Psychology*, Vol. 32 (2020), p. 73. https://doi.org/10.1016/j.copsyc.2019.07.008.
[58] The study by Haynie, et al. found that residential mobility was associated with a 60 percent increase in risk among adolescent girls. Dana L. Haynie, Scott J. South, Sunita Bose, 2006. "Residential mobility and attempted suicide among adolescents: an individual-level analysis." *The Sociological. Quarterly*, Vol. 47, No. 4, pp. 708–714. https://doi. org/10.1111/j.1533-8525.2006.00063.x.; Paulien Hagedoorn & Marco Helbich, "Longitudinal effects of physical and social neighbourhood change on suicide mortality: A full population cohort study among movers and non-movers in the Netherlands". *Social Science & Medicine*, Vol. 294 (2022), p. 2.
[59] Hagedoorn & Helbich, "Longitudinal effects of physical and social neighbourhood change on suicide mortality", p. 2.
[60] Hagedoorn & Helbich, "Longitudinal effects of physical and social neighbourhood change on suicide mortality", p. 3.

Regarding macro level variables, they are indicative of social disruption on an aggregated scale. Economic fluctuations are commonly used to measure anomie in longitudinal studies. As previously mentioned, Durkheim proposed a non-linear relationship between economic fluctuations and anomie. Times of prosperity should be considered as much a crisis as a financial crisis; both disturb the collective order.[61] The hypothesis is thus that sudden changes in economic conditions, regardless of direction, will be socially disruptive. In accounting for the historical aspects of Durkheim's theory, that a modernizing society would in and of itself disintegrate social relations, in-migration will be used as a proxy for aggregated levels of social disruption. Population growth and share of urban population has previously been used to some effect as proxies for urbanization and/or modernization. By using in-migration, the study captures the specific population growth that is associated with urbanization, as cities primarily grew through migration.[62] Results sprung from this variable should however be cautiously interpreted as the variable could potentially capture many things. Previous studies considering the historical perspective and using urbanizations proxies have measured overall suicide rates in entire countries, meaning that they have also captured rural areas. [63] As Anderson warned, suicide rates in the rural areas were higher than in industrial towns and inferring too much from long-lived ideas can be dangerous as in the present study, there is no data for rural areas.[64] Essentially, it becomes contradictory to claim that urbanization caused increasing suicide rate when only investigating urban areas. Either way, modernization is at the heart of Durkheim's thesis and should be controlled for, and although not a control variable in an empirical sense, it could be considered a theoretical control.

Durkheim postulated that poverty would protect against suicide via its restraining nature. If an individual is limited in the first place, their desires will not reach as far and in facing resistance, one has to rely on their community.[65] Research has instead shown the opposite to be true. Mainly through poverties association with other suicidogenic factors such as unemployment, alcoholism and illness.[66] In the absence of these measures, testing the theories of integration and anomie through individual economic indicators become problematic. Instead

[61] Durkheim, *Suicide*, pp. 206–207.
[62] Gustafson, "Industrialismens Storstad", pp. 32–33.
[63] Stack hypothesized a quadratic relationship between suicide rates and urbanization, where rates would rise due to the social disruption caused by the first waves of rural to urban migration. As new urban dwellers became adjusted, the rates would plateau and decline. Stack, "Suicide: A 15-Year Review of the Sociological Literature Part II", p. 165; Chen et al., "Socio-economic studies on suicide", pp. 287-288.
[64] Anderson, "Did Suicide Increase with Industrialization in Victorian England?", pp. 165-166, 172-173.
[65] Durkheim, *Suicide*, pp. 214-215.
[66] Stack, "Suicide: A 15-Year Review of the Sociological Literature Part I", p. 155.

Hamermesh & Soss theory of perceived utility is adopted into a hypothesis through use of age and, cautiously, individual social class.[67]

Given the complexity of the phenomena, establishing an all-encompassing theory of suicide may very well be a fruitless project. With this said, I believe that any theory regarding suicide should not seek to explain suicide. Rather, it should seek to explain what factors triggers an increased risk for suicide. Perhaps a minor, although important, distinction. The hypotheses formulated in table 1 are thus not assumed to be the *only* factors explaining suicide.

4. Material, data & variables

In 1878, due to the rapid urbanization in Stockholm, a new institution for censuses was established. To complement the church's registers, the city was divided into wards, or so-called *rotar*. Each rote had a *roteman* assigned to it. The roteman kept diligent notes on people and people's movement in Stockholm. In the rotemans's ledgers, information such as age, profession, household constellation and civil status can be found on people living in a specific property. Additionally, information is provided when a person moves in to said property and from which they moved.[68]

The roteman's ledgers provide plenty of information. They do not however provide information on cause of death. A growing city did not solely mean increasing administrative issues but also issues of health and sanitation. Thus, new institutions were formed whose goal was to give the cities political means to intervene when living conditions were deemed unsustainable.[69] Previously the priests of the parishes had been responsible for the reporting of deaths but in the middle of century, this duty was overtaken by the doctors in the cities.[70] By 1878, the health committee had reformed once more and this time went by the name of *Hälsovårdsnämnden*. This new committee administered death certificates from the year of their founding until 1926, active during the same period as Rotemannen were. These certificates contained information on first- and last name as well as primary and secondary cause of death.[71] As part of a digitization efforts made by the city archive and SwedPop, individuals have been

[67] HISCLASS does not directly take into account income in the classification scheme. However, the different social classes can be said to roughly represent historical earning power. Hamermesh, & Soss, "An Economic Theory of Suicide", pp. 84–86; van Leeuwen & Maas, *HISCLASS: A Historical International Social Class Scheme*, pp. 26-27.
[68] Geschwind and Fogelvik, *The Stockholm Historical Database*, pp. 207-210.
[69] Burström, Bo, Gloria Macassa, Lisa Öberg, Eva Bernhardt, and Lars Smedman, "Barnadödlighet, fattigdom och sanitära reformer: Stockholm 1878–1925", *Socialmedicinsk tidskrift*, Vol. 80, No. 3, (2003), p. 210.
[70] Thom Axelsson. *Självmord i teorin : En analys av olika forskares tolkning av självmord*. (Linköping University Electronic Press, 2000), p. 20. https://urn.kb.se/resolve?urn=urn:nbn:se:liu:diva-28687.
[71] *Stadsläkarens dödsbevis | Hitta i arkiven | Stockholms Stad*.

identified across both datasets and linked using an identification code.[72] In merging these databases, the thesis will be provided plenty of information on an individual level. The following chapter will outline on what premises the study extracted data on suicides from the material and further on, how variables have been extracted from the data.

4.1. Data

The data to be used in the analysis consists of 4.5 million observations between the years of 1878 and 1925 covering people of the age 14 and above. Each observation represents a citizen of Stockholm, and some individuals may have more observations than others. In turn, 3159 of an individual's last observation has been linked toward a cause of death stated as suicide. The coding of suicides has for the most part been done in present study. For this reason, there are a few things to be mindful of.

As of 1864, suicide was decriminalized in Sweden. However, the right to be buried in Christian ground was still denied those who had committed suicide until 1908.[73] The historical underreporting of suicide is well acknowledged problem among studies considering longer time periods.[74] The increasing suicide rates during 19th century is often explained by a greater tendency to report suicides than have been before. [75] On the other hand, some argue that the reporting of suicides could be even less reliable in the hands of the doctors in mid-19th century, due to the emergence of life insurance.[76] If the case of the latter, then the suicide rates rose despite of underreporting. Whichever the case, the observable rates rose substantially during the period. It should be acknowledged that the definition of suicide has varied over time and what constitutes a suicide can be ambiguous, which could also impact reporting. During the 18th century in Sweden, it was not uncommon to classify death by duel as a form of suicide for example. Furthermore, if provoking the law, or any other instance where death is a possible consequence, with the intent of dying, would this not constitute a suicide? In the eyes of God

[72] SwedPop. (2026) Version 4 of Harmonised Data in SwedPop. (Version 4). Original database: Rotemannen database, SE/SSA/0032/03 & SE/SSA/0561/01/01, Stockholms Stadsarkiv. Demographic Data Base (DDB), Umeå University. https://doi.org/10.48524/SWEDPOP-V4-0.
[73] Thom Axelsson. *Självmord i teorin : En analys av olika forskares tolkning av självmord*. (Linköping University Electronic Press, 2000), p. 27. https://urn.kb.se/resolve?urn=urn:nbn:se:liu:diva-28687.
[74] Social constructionists have mostly raised the issue of causation from sources where the empiricism builds on missing data. Suicide rates should in turn be treated as social constructions, due to their close association with the cultural contexts in which they are constructed. Pescosolido remarks that, while valid critique, it has not resulted in deductive theory. Pescosolido, "Bringing Durkheim into the Twenty-First Century", pp. 268-269.
[75] To be noted is that observable suicide rates were exceedingly low during 1600-1700, ranging from 0 to 5 per 100 000 population during the 17th century. Arne Jansson, *From Swords to Sorrow: Homicide and Suicide in Early Modern Stockholm*, (Almqvist & Wiksell International 1998), pp. 25-26.
[76] Axelsson, *Självmord i teorin*, p. 20.

and in turn the law, this does not seem to be the case thus a coroner's report would not reflect this.[77]

In coding suicides, the present study was to that end limited by the coroner's definition of suicide, that is to say when 'suicide' is explicitly stated as cause of death. The coroner's reports are quite descriptive when it comes to specific approaches. By the looks of it, there is no suicide by duelling for example, a practice that was also banned at the time. Regarding the issue of accidents, there are a large number of mortalities by drowning in the dataset. Some of these seem to be confirmed suicides. As for the rest, it is disputable for both the coroner and for us, to determine if a case is suicide or accident. With this in mind, there could supposedly be a number of hidden suicides occurring during the time.

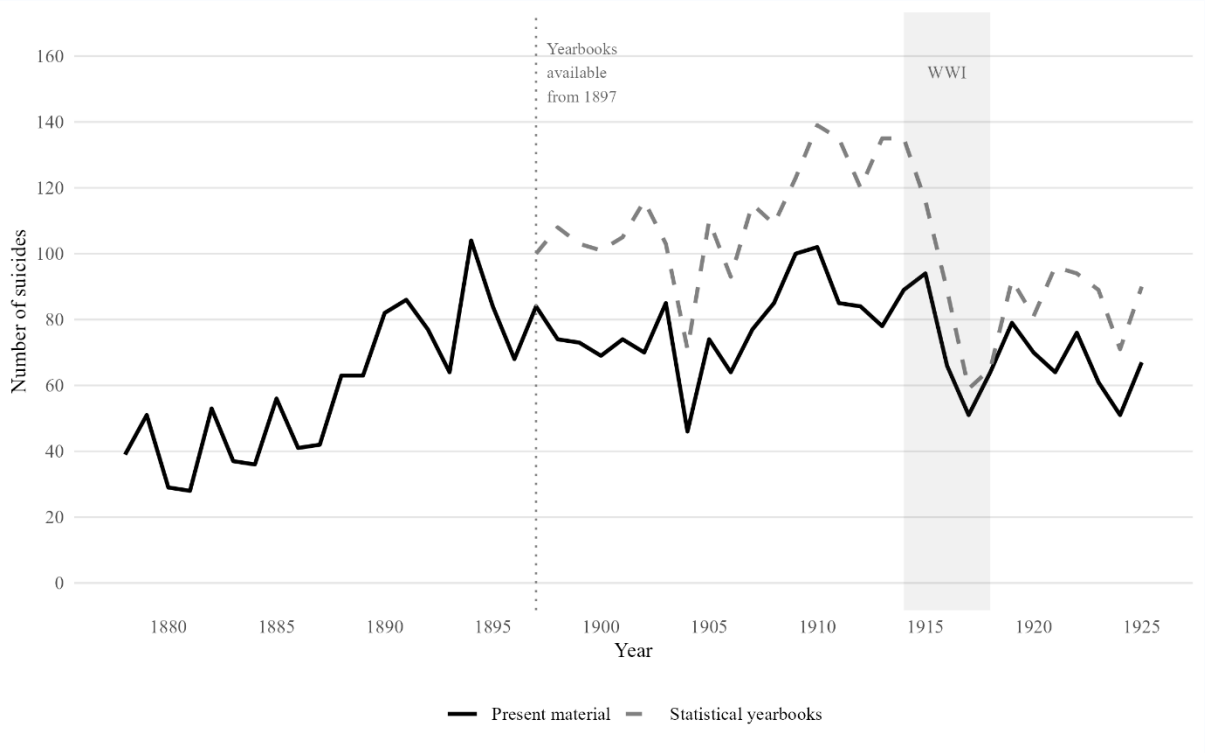
With all this said, there is a discrepancy between the official statistics of Stockholm from the time and the material at hand. Figure 1 shows how many suicides occurred during the period of interest according to the official statistics of Stockholm and according to the due material. The discrepancy is due to two reasons mainly. One, due to the nature of the combination between Rotemannen and hälsovårdsnämnden, there will simply be a loss of data. This is, however, completely at random and should not impact on the reliability of the data. The posts in Rotemannen have been matched with the posts in hälsovårdsnämnden given that Rotemannen had registered a person as deceased. If for any reason, clerical or whatnot, an individual has not been registered as D then the match will not happen and then the data is lost. In the current material there are 211 267 posts, meaning that this many causes of death were able to be matched. This is compared to the official statistics of Stockholm in which 242 186 were registered as dead between the years 1878 and 1925.[78] This means that we can expect an approximate 13% discrepancy in suicide due to random chance.

[77] In early modern Stockholm, this seemed to be somewhat of a common practice. Arne Jansson found in an investigation of trial records the specific phenomena of *suicidal murders*, wherein a person, most often a woman, murders another person, most often a child, with the intention of being executed. Jansson*, From Swords to Sorrow*, pp. 1-3; Axelsson, *Självmord i teorin*, 1–326.

[78] Statistisk årsbok för Stockholms Stad, "Befolkning, sundhet och dödlighet" (Stadens statistiska kontor, 1904–1929)

Figure 1: Annual Suicide Counts, Stockholm 1878-1925

Present material (hälsovårdsnämnden) vs. Statistical Yearbooks of Stockholm



Statistical yearbooks first published in 1904, with retrospective data from 1897.
Shaded area = First World War (1914-1918)

The second reason is somewhat more damning. Causes of death were not standardized at the time. When someone passed away due to suicide, the cause of the death could range from simply "suicide" to "jumped from the fourth floor" to "gunshot wound", where in the case of the last it is only specified in the secondary cause of death that the gunshot was fired by the deceased themselves. Attempts have been made by the digitization project to standardize the causes according to the coroner's notes. However, these have only included the explicitly stated suicides, and some of the implicitly stated. These add up to 1314 suicides during the entire period of interest, while the data from the yearbooks amounts to 2963 suicides just for the period 1897 to 1925. A far greater discrepancy than indicated when comparing the total mortality between the yearbooks and the data. It is possible to extract more suicides from the data, but it will require some sleuthing.

The statistical yearbooks categorize suicide, homicides and accidents as *violent deaths*. Suicides are in turn counted separately from other violent deaths and not only by the number of suicides per year but furthermore the amount that committed suicide by firearms, sharp tools, drowning, hanging, poison or otherwise. Violent deaths by gunshots wounds and stabbing wounds, excluding suicide, happen around 1-5 times per year during the period. Meanwhile,

as method of suicide they happen 60 to 20 times a year respectively. By comparing the individual causes of death to the specific numbers in the yearbook it is possible to get a much closer estimate. To exemplify; in 1911, 39 people committed suicide by firearms according to the yearbooks at the same time as only 2 other people died to gunshot wounds while not being suicide.[79] The same year in Rotemannen, there were 2 murders by gunshots thus one can assume that the rest of the gunshot deaths are suicides, even though not explicitly stated by the coroner. Although not perfect, it does provide a more reliable way of getting a closer estimation. In the end, it becomes a question of identifying and excluding homicides and accidents rather than identifying suicides. Discrepancy considered, the data gathered for the sake of the analysis seem to follow the trend of the year-books data when considering figure 1, a good sign.

Table 2: Suicide by Method: Present Material vs Official Statistics

Stockholm 1897–1925

	Present material		Statistical yearbooks		
Method	N	%	N	%	Diff. (%pts)
Hanging	847	39.3	1118	37.7	1.6
Firearms	506	23.5	730	24.6	-1.1
Poison	188	8.7	286	9.7	-1.0
Drowning	145	6.7	492	16.6	-9.9
Sharp tools	81	3.8	111	3.7	0.1
Crushing	57	2.6	183	6.2	-3.6
Other	20	0.9	43	1.5	-0.6
Unspecified	*312*	*14.5*	—	—	—
Total	**2156**	**100.0**	**2963**	**100.0**	**—**

Difference = present material % minus yearbook %. Positive values indicate overrepresentation in material relative to official statistics. Total for present material concerns comparable years to the yearbooks, not the total amount to be used in analysis.

In table 2, the official distribution of suicides per cause from the yearbooks are compared to the distribution in the present material, for the years 1897 to 1925.[80] There is a ten to fifteen percent discrepancy across the board, aligning with the total discrepancy when looking at all-cause. Notable exceptions are of course crushing and especially drowning for which distribution percentage differs by 3.6 and 9.9 respectively. It is, to some extent, to be expected

[79] Statistisk årsbok för Stockholms Stad, ”Befolkning, sundhet och dödlighet” (Stadens statistiska kontor, 1916)
[80] As mentioned in the comments of figure 1, the yearbook was first published in 1904, with some retrospective data available. Table 2 thus only entails distribution between 1897 and 1925 for both the present material and the yearbooks. Statistisk årsbok för Stockholms Stad, ”Befolkning, sundhet och dödlighet” (Stadens statistiska kontor, 1904–1929)

that these causes differ as in these cases it is hard to determine intent.[81] However, since the yearbooks present a much higher total number, the issue must lie in the registration. There is unfortunately no easy fix for this. Choice of method can be dependent on access to means, which in turn can be impacted by factors such as occupation or income, thus it is an important data quality check.[82] Method have also been shown to vary between gender, which furthermore indicates the need to not only have accurate yearly distribution but also accurate distributions of method.[83] Part of the discrepancy in table 2 may be alleviated by the 312 unspecified suicides. By simultaneously having accurate numbers for other causes, it should still provide a faithful representation.

Potentially, the deaths that were not registered as such in Rotemannen, may be strongly correlated with one of the explaining variables to be used. Since residential mobility is of interest, a death might have gone unregistered for specifically reasons of residential mobility. While it is not possible to say, it should be kept in mind. In the end, clerical errors and specificity of coroners make the foundations of the selection. Errors in regard to the yearbooks seem to be consistent between years, gender and specific causes of suicide. That is to say; the sample should for the most part be regarded as unbiased and should still be suitable for analysis.

4.2 Variables

The following section will introduce the measures to be used in the analysis, how they have been collected and if adjusted, how they have been adjusted. All individual level measures are collected from Rotemannen so first these are introduced alongside a brief run-through of how the data in the set is structured. Afterwards, macro level measures are introduced, where most are collected from external sources, with an exception. Table 1 presents a quick overview of these variables. For the individual level variables, number and distribution of spells are showcased alongside number and distribution of suicides with yearly suicide rate per 100k population. Spells refers to the number of observations in which time is measurable. This will

[81] Axelsson, *Självmord i teorin*, p. 20.

[82] Stack brings up the example of dentists as having a markedly higher suicide rate than other high-status occupations. He explains this has dentists being seen as lower status within the medical community and having strained relationships with their patients, due to not many people enjoying going to the dentist. Most assuredly, it should furthermore be possible to explain this through an access of means. Stack, "Suicide: a 15-Year Review of the Sociological Literature, part I", p. 156.

[83] A study of suicide methods showed that in the latter half of the 20th century Västerbotten, firearms were favored by the male population while poisons and drownings were favored by the female. Jayanti Chotai, Ellinor Salander Renberg, & Lars Jacobsson, "Method of Suicide in Relation to Some Sociodemographic Variables in Northern Sweden". *Archives of Suicide Research*, Vol. 6, No. 2 (2002), pp. 114–115. https://doi.org/10.1080/13811110208951169; See also Kushner, "Women and Suicide in Historical Perspective", pp. 546-551.

be discussed further on in the methodology section. For macro-level variables, mean, standard deviation and range alongside number of observations are showcased. Sources from which data has been gathered are included for all variables of interest.

Table 3: Variable Overview

Stockholm 1878–1925

Variable	Description	Source	Distribution / Coverage
Individual-level variables			
Suicide	Binary outcome: death recorded as suicide in death register	Hälsovårdsnämndens death register, linked to Rotemannen via post ID	3,159 suicides / 4,560,503 observations
Sex	Male or female	Rotemannen population register	Male: 43.7% / Female: 56.3%
Age group	Age at observation start, grouped as 14–29, 30–49, and 50+	Rotemannen population register (derived from birth date)	14–29: 52.4% / 30–49: 34.4% / 50+: 13.2%
Birthplace	Birthplace parish categorised as Stockholm born, internal migrant, foreign born, or unknown	Rotemannen population register	Internal migrant: 62.5% / Stockholm born: 34.2%/
Household size	Number of co-residents in household: Alone (ref.), Small (1–4), Medium (5–9), Large/Institutional (10+)	Rotemannen population register (HF household overlap)	Alone: 49.6% / Small: 37.5%
Residential mobility	Annual residential moves per year of Stockholm observation, log-transformed as log(1 + moves/year). Implicit baseline: zero moves per year	Rotemannen population register (genuine address changes, counting process formulation)	Mean: 0.31 / Median: 0.21 moves/yr
Social class	HISCLASS occupational class: Unskilled workers (ref.), Lower skilled workers, Skilled workers, Lower managers and professionals, Higher managers and professionals, Farmers and farmworkers	Rotemannen (occupation), HISCO/HISCLASS classification	Coverage: 92.6% of observations
Macro-level variables			
In-migration rate	Annual in-migrants per 1,000 population, Stockholm municipality	SCB historical yearbooks	Mean: 67.1 / Range: 13.1–96.7
Nuptiality rate	Annual marriages per 1,000 population, Stockholm municipality	SCB historical yearbooks	Mean: 8.7 / Range: 7.1–9.8
Real wage cycle	HP-filtered cyclical component of real wage index for unskilled workers (λ=100)	Prado's real wage series (2010); own HP filter	Mean: -0.03 / SD: 6.12
GDP growth	Annual real GDP volume growth rate, Sweden	Edvinsson's GDP growth series (2014)	Mean: 2.36% / Range: -8.15–8.05%

Residential mobility log-transformed as log(1 + moves/year) to address right skew. HP filter applied with λ=100 (Hodrick & Prescott 1997). GDP growth from Swedish Riksbank historical database.

Missing distributions from table concerns birthplace and household size. Birthplace: Foreign born: 2.7% / Unknown: 0.6%. Household size: Medium (5-9): 10.1% / Large/institutional: 2.8%.

4.2.1. Micro

The data from Rotemannen is ordered around posts. An individual's first post is created when an individual either moved into Stockholm, was born in Stockholm or was already living in Stockholm at the time of the dataset's inception. Additional posts per individual are created when they move to Stockholm or are reregistered or if they exit the data, either by moving out of the city or passing away. One post contains information on identifiers such as name, codes, birthdate. It also contains address, parish, birthplace, location from where they moved, location to which they moved and whether or not person enters or exits the data. In-codes and out-codes respectively, contain information on what category of moves triggered the post. For example, if they move from or to another county in Sweden or within Stockholm. They are the same with the exception that F (born) is not in present among the out-codes and D (death) is not present among the in-codes.

There are three identifying codes which can be used to find individuals, a post ID (RoteID), an individual ID and a HF (head of household) ID.[84] The *post ID* is unique for every post and in turn row. The post ID code for every post with an out code of value D is also found in the combined digitization of the data from hälsovårdsnämnden. Through this it was possible to give every post coded D a specific cause of death. *Individual ID* is unique for every individual, since an individual could appear several times in the data if they moved or were for some reason re-registered at some point. The HF ID for each post matches the post ID of the HF at the same point in time. See table 4, where structure of data is exemplified.

Table 4: Example of Data Structure in Rotemannen

Post ID	Last name	First name	Individual ID	HF ID
0000001	Andersson	Pelle	001002	0000001
0000002	Andersson	Karin	002003	0000001
0000003	Andersson	Pelle	001002	0000003
0000004	Andersson	Karin	002003	0000003

Some variables excluded due to not being used in analysis. PERSON 1 (Pelle, post 0000001) is the HF of PERSON 2 (Karin) in the first post. In the second post (posts 0000003-0000004), Pelle's new post ID becomes the new HF ID.

Social class was based upon occupation title. As mentioned, Rotemannen contains information on occupation for almost all individuals across each post. The digitization effort has in turn assigned each post containing information on occupation, a code according to the Historical International Standard Classification of Occupations (HISCO). For the sake of the

[84] Geschwind & Fogelvik, "The Stockholm Historical Database", pp. 216-225.

analysis, the HISCO-codes have subsequently been re-coded according to the Historical International Standard Class Scheme (HISCLASS). HISCLASS is a 12-step classification scheme based upon skill level, degree of supervision, whether manual or nonmanual and whether rural or urban. All posts were then assigned to a social class dependent on their HISCLASS code.[85] For this analysis, I used a simpler 7-step scheme, it was as follows:

1. Higher managers and professionals (1 – 2)
2. Lower managers and professionals (3 – 5)
3. Skilled workers (6 – 7)
4. Lower skilled workers (9)
5. Unskilled workers (11)
6. Farmers and fishermen (8)
7. Farmworkers (10, 12)
8. NA

66% of men and 45% of women had their own occupational code. A substantial number of posts had the simple occupation title *worker* or synonymous title. These did not have their own HISCO-code due to not being able to classify industry. For the sake of the analysis, these were coded as Unskilled workers. If a post still was missing a HISCO-code, they were matched to their HF's code. Women were assumed to have the same class as their partner, if they had one and were missing code. Similarly, children without occupation were assumed to have the same class as their father. After imputation, 92.6 percent of posts had been assigned a social class. Farmers and Fishermen were ultimately merged with Farmworkers due to exceedingly low amounts of suicide in the latter group. Both categories in general are problematic due to the former being a heterogeneous group, ranging from large to small scale farm owners and both groups underrepresented when observing urban areas.[86]

The *Birthplace* variable is constant across spells for unique individuals. It uses the county code for counties in Sweden, where 1 or 2 means an individual was labelled *Stockholm born* and 50 means an individual was labelled *Foreign migrant*. Any value in between means the

[85] van Leeuwen & Maas, *HISCLASS: A Historical International Social Class Scheme*, p. 57.
[86] Martin Dribe, Therese Nilsson & Anna Tegunimataka, "Urban Lives: A Micro-Level Approach to Economic and Demographic Change in the Twentieth Century" in *Urban Lives: An Industrial City and Its People During the Twentieth Century*, Martin Dribe, Therese Nilsson & Anna Tegunimataka (eds), (Oxford Academic, 2024), p. 16. https://doi.org/10.1093/oso/9780197761090.001.0001

individual was labelled as *Internal migrant*. Those who were missing a county code were labelled as *Unknown*. *Internal migrants* were used as the reference group.

Residential mobility is a continuous variable, referencing moves per year. The variable was log-transformed due to extreme outliers.[87] To account for those who never moved, log estimates were defined as logged value + 1. It was modelled continuously with zero moves per year as the implicit baseline. In interpreting the results, a one unit increase in log value is equal to moving 1.7 times a year. Residential mobility must be treated somewhat cautiously. In an actively expanding city, the decision to move might not be a decision at all. Furthermore, there can be confounding factors behind a move as well, such as employment status, change in household size or economic status. Household size and social class is possible to control for, but there are limitations in controlling for whether someone was employed or not.

4.2.2. Macro

The macro variables have mostly been collected through external sources with the exception of *in-migration*. By separating those entering the Rotemannen-data through birth and those entering through migration, it is possible to produce annual data on population growth through specifically migration. The distinction is mainly important for two reasons. Population growth and/or density are often used as a proxy for urbanization and modernization.[88] During times of industrialization, fertility rates often decline, a fact also observable for the current period.[89] If in-migration is the primary driver of urbanization during this period rather than natural population growth, as the fertility data suggest, then in-migration rate should more accurately capture the disorienting effects of associated with rapid urban growth. Furthermore, migration would reasonably represent a more emergent anomic effect as competition for housing and employment increases. The final rate used in the analysis is total in-migrants per 1000 of population per year.[90]

The other two aggregated indicators are of industrial activity; they are *Gross Domestic Product* (GDP) and *real wage change* for Sweden. The study of economic fluctuations effect on suicide rates traces back to Durkheim. Studies empirically testing this usually make use of GDP growth and unemployment rates. Unfortunately, reliable data on unemployment are not

[87] Distribution in moves per year across the entire population was Min: 0, Median: 0.21, Mean: 0.31, Max: 44.
[88] Chen et al., "Socio-economic studies on suicide", pp. 287-288.
[89] Luciana Quaranta & Maria Stanfors, "The Gender Revolution: Marriage, Fertility and Divorce in the Industrial City", pp. 189-191.
[90] Total population was collected from the statistical yearbooks of Stockholm. Statistisk årsbok för Stockholms Stad, "Befolkning, sundhet och dödlighet" (Stadens statistiska kontor, 1904–1929)

available in Sweden for this time, except for the latter third of the study period. GDP growth has been collected from Rodney Edvinssons reconstruction of the Swedish historical national account (SHNA).[91]

Real wages on the other hand are often used as an indicator of living conditions. The particular series used in the analysis, stretching between the years 1860 to 2007, was constructed by Svante Prado. It covers the real wages of male and female manufacturing workers across nine industries.[92] For the pre-1913 period, Prado uses the series found in Wages in Sweden by Gösta Bagge, et al. with a few modifications to also account for the quarrying industry and the chemical industry. For the post-1913 period, Prado uses official statistics published by first the Social Board and then later Statistics Sweden on average hourly wages by industry.[93] While both indicators refer to the entirety of Sweden rather than Stockholm, the city generally followed the trend of the country in terms of wage growth and, due to size of Stockholm's industry, GDP fluctuations.[94]

For real wage change the Hodrick-Prescott (HP) filter has been employed with a conventional smoothing parameter of $\lambda = 100$. The filter decomposes a series into a long-run trend component and a short-run cyclical component. The cyclical components, which represent deviations from trend, are used as the explanatory variable in the model.[95] Deviation from trend is preferred over other measures, such as changes in percentage per year, since it captures the anomic dimensions of economic fluctuations as postulated by Durkheim.[96] For industrial employment specifically, the filter was applied to the natural logarithm, thus yielding deviations expressed in approximate percentage.

[91] Rodney Edvinsson, "The Gross Domestic Product of Sweden within present borders, 1620–2012" in *Volume II: House Prices, Stock Returns, National Accounts, and the Riksbank Balance Sheet, 1620–2012*, Rodney Edvinsson, Tor Jacobsson & Daniel Waldenström (Ekerlids Förlag & Sveriges Riksbank, 2014), pp. 138, 145.
[92] Svante Prado, "Nominal and real wages of manufacturing workers, 1860–2007" in *Exchange Rates, Prices, and Wages, 1277-2008*, Rodney Edvinsson, Tor Jacobsson & Daniel Waldenström (Ekerlids Förlag & Sveriges Riksbank, 2010), p. 504.
[93] Prado,"Nominal and real wages of manufacturing workers, 1860–2007", pp. 484-489, 504.
[94] Data was collected on industrial employment in Stockholm and used in earlier models with minor significant results. Decision was made to use GPD as economic indicator instead as industrial employment capture only a specific part of the economy, thus GDP should be more representative. Furthermore, GDP should roughly correlate with trends in unemployment as well, an important predictor of suicide rates.
[95] Robert J. Hodrick & Edward C. Prescott, "Postwar U.S. Business Cycles: An Empirical Investigation", *Journal of Money, Credit and Banking*, Vol. 29, No. 1 (1997), pp. 3-5. https://doi.org/10.2307/2953682
[96] This meaning that any fluctuation, independent of direction, is cause for anomie. Durkheim, *Suicide*, pp. 202-206.

4.2.3. Controls

Suicide has strong associations with several factors and these need to be controlled for in the analysis. The most major ones are *age* and *gender*, which consistently have been shown to correlate with suicide. However, additional controls are made for social factors not necessarily associated with urbanization, in this case *household size* and *nuptiality*.

Age is somewhat accounted for by choice of main model, which will be elaborated on in the methodology section. To further control for compositions, age groups are included as covariates in the models. Age groups were defined as 14-29, 30-49 and 50+ in accordance to other empirical studies on suicide considering longer time perspectives, although somewhat rougher due to limited suicide observations in dataset.[97]

As for gender, Durkheim explained the difference as women being less involved in the social world than men, and thus less subject to the effects of it. Instead, women's immersion in the traditional family life would protect against suicide, since suicide was a consequence of collective existence.[98] Cases of suicide by women could be dismissed as individual isolated actions of emotional distress. Women have since been shown to be overrepresented in statistics on suicide attempts, a phenomenon on which Durkheim merely makes passing comments on. As previously mentioned, the study will not be able to account for suicide attempts. The main implication of this is that women will be underrepresented as male suicide rates are significantly higher. Howard I. Kushner argues for the importance of treating attempts the same way as completed suicides as, when aggregated, completed suicides and attempts show no difference in behaviour. In addressing suicide as a male issue, assumptions of female immunity to suicide will continue to be reinforced. The difference in historical data, he argues, is due to cultural differences in access to methods. Historically, women have had less access to more lethal means, such as firearms, favouring instead drowning, hanging or poison.[99]

As a proxy and control for social isolation, *household size* is used. *Household size* was calculated by using the HF ID. By extracting the HF-code for an individual post it was possible to calculate how many household members an individual had during a specific spell. Due to a number of extreme outliers in the data, household size is divided into five categories. *Living*

[97] Sensitivity check was additionally performed to account for other commonly used age groups in historical demographic studies as 14-24, 25-44. 45-64. 65+. Checks found no substantial heterogeneity in groups, but 65+ group had insufficient amounts for reliable estimations. Thibodeau & Lachaud, "Impact of economic fluctuations on suicide mortality in Canada (1926-2008)", p. 307; Hedefalk, Svensson & Tegunimataka "Migration in Times of Economic Growth and Recession", p. 132.

[98] She is however not immune to committing murders of other natures, Durkheim points out. Infanticide, abortions and domestic murders, are instead within her sphere of influence, the household. Durkheim, *Suicide*, pp. 307-308.

[99] Kushner, "Women and Suicide in Historical Perspective.", pp. 546–552.

alone (0), *small household* (1-4), *medium household* (5-9) and *large/institutional household* (10+) with those *living alone* being the reference group. Categories are arbitrary and have thus been sensitivity tested, although results are not reported. No substantial differences were recorded when testing for other intervals.[100] Institutional households often contain several hundreds of members at a time. Seemingly these are poorhouses or some other kind of aid-relief.

Finally, interpersonal relationships should be accounted for on some level. Durkheim theorized that divorce should be one of the main drivers of anomie and succeeding literature has shown this to be a relevant relationship.[101] Due to lower divorce frequency during the period, divorce rates may be quite low and thus misleading. As a proxy, the study will use *nuptiality rates* counted as the number of marriages entered per year per 1000 of population. The rationale behind using nuptiality instead is that it should have the opposite effect, essentially being a protective factor. It is thus included as a control for the effects of marital life upon suicide. To be noted is that it was not uncommon for citizens of Stockholm to live together outside of marriage albeit still operating under similar conditions. This phenomenon is known as *Stockholmsäktenskap*.[102] The data at hand cannot account for these unfortunately, so the nuptiality rate will thus not be able to capture all instances of family formations.

5. Methodology

The study employs two survival models for the analysis. The main model to be used is Cox proportional hazards model and for robustness testing, Fine & Gray models are used. Survival analysis entails modelling time-to-event data to describe the length of time until event of interest occurs. It does this through so-called *spells*. A spell contains information on how long a person exists within specific states, counted as person-years, until A, starting a new spell or B, reaches an end event.[103] In the case of the study at hand, there are three types of events. An individual can experience the following: 0, living (in Stockholm), 1, exits data through suicide

[100] Intervals tested were: 1-3, 4-8, 9+ and 1-5, 6-10, 11+.
[101] Durkheim, *Suicide*, p. 237; Stack, "Suicide: a 15-Year Review of the Sociological Literature, part II", pp. 167-168.
[102] While seemingly frequent, the Stockholm marriage was also frowned upon and not a legalized form of partnership. Data is thus hard to come by. Margareta R. Matović, "Stockholmsäktenskap: familjebildning och partnerval i Stockholm 1850–1890" Monografier / Industriarbetarklassens uppkomst: Stockholms arbetare 1850–1914, 57. Diss., Stockholm University, 1984, pp. 141–144.
[103] Structure of survival analysis may depend on subject of interest. In present case, time is from birth in Stockholm or move into Stockholm until exit. In investigating cancer mortality, time may be from diagnosis until to death. It is also possible that the event of interest may occur several times. For example, in the case of specific infections due to disease. Terry M. Therneau & Patricia M. Grambsch, *Modeling Survival Data: Extending the Cox model*, (Springer: New York 2000), p. 4.

and 2, exits the data through other cause (moving away from Stockholm or dying to another cause). In this case, the spells in turn are the time between posts. To exemplify; Person A entered the data through moving to Stockholm in 1885 on the 29th of April, currently living with three other household members. In 1894 on the 22nd of August, person A moved to a new address, now only living with one other household member. Spell duration is in this case 3402 days. On the 4th of March 1896, person A dies due to pneumonia. Person A was then censored from the data. There are a variety of survival models based upon basic premise described here, and each have their advantages and disadvantages. The following chapter will present which models have been used in the analysis and the rationale behind the choice of these.

5.1. Model specifications

The primary method was the extended Cox proportional hazard model as formulated by Therneau and Grambsch.[104] Choice of method in this case is largely due to data structure but the Cox model is also one of the most common ways of analysing survival data. The Cox model allows for age to be used as a time scale instead of calendar time. Whereas using calendar time and including age as a covariate might produce misleading results, this makes it so that age specific baseline hazards are implicitly controlled for.[105] An important feature given the association between suicide and age.

The counting process formulation of the extended Cox model includes an interval structure to account for breaks in observation, expressed as age_start and age_stop.[106] The data at hand contains left truncation, meaning that for subjects who have a delayed entry into the data are unobservable for periods albeit still reasonably being at risk. Since Stockholm is expanding primarily through in-migration, there will be several cases of time not being observable. Take the case of person A once again, which moved to Stockholm in 1885. The study period begins in 1878 and thus, person A is not observable until 1885, while still at risk for suicide. Left truncation also applies to all individuals living in Stockholm at the time of period start.[107] It furthermore naturally handles right-censoring, people who exit the data for other reasons than suicide.[108] Given the high mobility in Stockholm during this period, many people were moving out of the city as well. The soon-to-be married couple from Dalarna might

[104] R-Package used for Cox model was "survival". Terry M. Therneau, "_A Package for Survival Analysis in R_." R package version 3.8-3 (2024), https://CRAN.R-project.org/package=survival
[105] Therneau & Grambsch, *Modeling Survival Data: Extending the Cox model*, pp. 39-40, 75.
[106] Therneau & Grambsch, *Modeling Survival Data: Extending the Cox model*, pp. 68-69.
[107] Therneau & Grambsch, *Modeling Survival Data: Extending the Cox model*, p. 75.
[108] Therneau & Grambsch, *Modeling Survival Data: Extending the Cox model*, p. 4.

have just stayed in Stockholm for a year or two, saving up for their marriage. The interval formulation in the counting extension of the Cox model allows for each subject to contribute to the risk set only during periods of observation. Since each individual is subject to time-varying covariates, such household size and history of residential mobility, this is even more important.

A Cox model produces estimates on hazard ratios (HR), which should be interpreted as relative risk compared to a baseline or reference. The Cox model assumes proportional hazards (PH), meaning that the HR between two subjects remains constant.[109] The HR's were assessed by use of Schoenfeld residuals, as proposed by Therneau and Grambsch. If PH assumption was violated, covariates were stratified. Stratification allows separate baseline hazards to be estimated for each stratum while retaining shared covariate effects across strata. This means that the stratified variable in question is controlled for but is not directly quantified.[110] Sex was stratified for the main models given the well documented differences in suicide rates between men and women.[111] HR's for independent covariates are thus assumed to be the same across sex for the sake of the model.[112]

Another assumption of the ordinary Cox model is that each observation will be independent. This is not the case in the current data since each individual more often than not contributes with multiple spells. To account for within-subject correlation, robust sandwich standard errors clustered on individual identifiers are used.[113] Additionally, to handle event ties, when events occur at the same time, these were handled through Efron approximation, as per Therneau and Grambsch's recommendation.[114]

5.2. Robustness tests

For robustness testing, the analysis compared the results of the Cox models against a Fine & Gray model, which operates under different assumptions. The Cox model regards deaths due

[109] Therneau & Grambsch, *Modeling Survival Data: Extending the Cox model*, pp. 127-128.
[110] Therneau & Grambsch, *Modeling Survival Data: Extending the Cox model*, pp. 44-46.
[111] Hagedoorn & Helbich, "Longitudinal effects of physical and social neighbourhood change on suicide mortality", p. 4.
[112] Sensitivity checks with models stratified on age groups and gender were made (see appendix B.1 and B.2). Models performed similarly as main models, with few distinctions. Results of these discussed further in "Results" chapter.
[113] Robust standard errors are handled natively in the "survival" package in R through the "cluster()" argument in "coxph()". Therneau T, "_A Package for Survival Analysis in R_." R package version 3.8-3 (2024), https://CRAN.R-project.org/package=survival; Therneau & Grambsch, *Modeling Survival Data: Extending the Cox model*, pp. 175-181.
[114] Efron is preferred over Breslow method when there are a moderate number of ties. Therneau & Grambsch, *Modeling Survival Data: Extending the Cox model*, p. 48.

to other causes than suicide, as censored events. What this does implicitly is assuming that competing risk is uninformative of the suicide risk. Suicide and other causes of death may have many common denominators in terms of risk, such as alcoholism, medical illness and social isolation.[115] If this assumption is violated then the hazard ratios in the Cox model might be upwardly biased. Fine & Gray accounts for competing risk in not censoring events that differ from the event of interest, thus allowing them to contribute to the risk-set, although with reduced weight.[116] It does this by directly modelling the cumulative incidence function.

In practice, employing this model allows for illustrating interactions between suicide events and other events. In studying cause-specific mortality, evaluation of risk is actively competing with other causes of death than suicide.[117] In the case of the Cox model, person A passes away from cardiopulmonary disease thus being censored due to not having committed suicide. According to the Cox model, person A is no longer at risk for suicide and does not provide information on the hazard of suicide. In reality, person A might have been at a very real risk for suicide, if not even due to the fact that they died to other causes before. As for interpretation, if the results between the models are similar then they should be regarded as robust. Estimates of the F&G models are presented as subdistribution hazard ratios (SHR), and should as with the Cox models HR, be interpreted as relative risk compared to a baseline or reference group.

[115] Stack, "Suicide: A 15-Year Review of the Sociological Literature Part I", p. 155.
[116] Jason P. Fine & Robert J. Gray. "A Proportional Hazards Model for the Subdistribution of a Competing Risk." *Journal of the American Statistical Association*, Vol. 94, No. 446 (1999), pp. 500–501. https://doi.org/10.2307/2670170.
[117] Christiana Kartsonaki, "Survival analysis", *Diagnostic Histopathology*, Vol. 22, No 7 (2016), pp. 268-269. ISSN 1756-2317, https://doi.org/10.1016/j.mpdhp.2016.06.005.

6. Results

The following chapter presents the findings from the analysis. First, a descriptive analysis is performed looking at distribution and suicide rates between categories. Second, the main model stratified upon social class is presented, in table 6, to test the hypotheses formulated in table 1. Finally, robustness models are showcased in table 7 along with comments on the results of sensitivity checks using age group and sex stratification instead of social class.

6.1. Descriptive analysis

Table 5 display the distribution of suicide events and person-years between the different variables. First off, some comments regarding the values displayed in table 5: Male suicide seems to be 4 times higher than female; this is consistent with the common suicides rates found among males and females. Furthermore, this somewhat alleviates the previously mentioned issue of there being more reliable data for suicide by firearms than other methods.[118] Age is also producing expected suicide rates, low among younger people and increasing with age.[119] Distribution on period also display expected results, with the rate declining during WWI. Most egregious distribution wise, is the unknown category of the birthplace variable displays an unusually high suicide rate of 68,2 suicides per 100k per year. More than 4 times as high as the second highest values of Stockholm born and Internal migrants. When taking a look at these individuals, there seemed to be no apparent common denominator. This means that it unfortunately is hard to say whether it is a statistical anomaly or indicative of something else not visible through the data. With having these aspects controlled for, simply looking at the suicide rate distribution reveals some interesting results. Regarding the age groups, there seem to be a clear gradient skewed toward older age being more at risk for suicide. This would align with Hamermesh & Soss's theory of perceived utility; utility being lost as a person increases in age. However, if social class is in any way indicative of general income, perceived utility should be lower among unskilled workers. Unskilled workers seem to have a substantially lower rate than higher classes.[120]

[118] Kushner, "Suicide, Gender and Fear of Modernity", p. 467; Chotai, Renberg & Jacobsson, "Method of Suicide in Relation to Some Sociodemographic Variables in Northern Sweden", pp. 114-115.
[119] Stack, "Suicide: A 15-Year Review of the Sociological Literature Part II", pp. 165-166.
[120] Hamermesh and Soss found that higher income groups had lower suicide rate. Present results do indicate a lower rate among the highest social class, but this would not explain the high rate among the lower managers. Hamermesh, & Soss. "An Economic Theory of Suicide", pp. 87–88, 97.

Table 5: Distribution of Suicide Events and Person-Years

Stockholm 1878-1925

Category	Spells	Person-years	Suicides (N)	Rate per 100,000 PY
Total	4,560,503	20,076,008	3159	15.7
Sex				
Female	2,567,831	11,020,125	709	6.4
Male	1,992,672	9,055,884	2450	27.1
Age group				
14-29	2,102,821	7,678,680	779	10.1
30-49	1,377,764	4,943,232	1456	29.5
50+	529,516	2,018,414	795	39.4
NA	550,402	5,435,682	129	2.4
Birthplace				
Internal migrant	2,850,615	12,577,846	1959	15.6
Stockholm born	1,561,829	6,628,836	1037	15.6
Foreign born	122,797	738,782	74	10.0
Unknown	25,262	130,545	89	68.2
Household size				
Alone	2,263,726	9,508,249	1362	14.3
Small (1-4)	1,711,020	7,678,305	1367	17.8
Medium (5-9)	458,984	2,156,277	344	16.0
Large/Institutional (10+)	126,773	733,177	86	11.7
Mobility rate				
Never moved	947,430	8,070,882	459	5.7
Low (<0.25/yr)	1,594,631	7,037,976	1319	18.7
Moderate (0.25-0.5/yr)	1,170,402	3,362,612	877	26.1
High (0.5-1/yr)	663,118	1,307,487	440	33.7
Very high (1+/yr)	184,922	297,052	64	21.5
Social class				
Unskilled workers	1,073,779	4,246,335	551	13.0
Lower skilled workers	1,184,832	4,747,057	692	14.6
Skilled workers	814,199	3,459,558	696	20.1
Lower managers and professionals	804,980	3,807,308	764	20.1
Higher managers and professionals	224,192	1,371,065	196	14.3
Farmers and farmworkers	121,225	556,442	109	19.6
Period				
1878-1889	1,346,312	6,178,209	824	13.3
1890-1899	892,870	4,322,493	824	19.1
1900-1913	1,389,008	6,610,398	1106	16.7
1914-1918 (WWI)	464,587	2,026,278	283	14.0
1919-1925	467,726	938,631	122	13.0

Person-years calculated from spell duration. Mobility rate = moves per year of Stockholm observation, categorised for descriptive purposes. The continuous log-transformed measure is used in the regression analysis.

NA age group (N=550,402 spells) reflects individuals outside the 14–50+ range or with missing birth dates. Excluded from regression models.

Discounting higher managers and farmers, social class seem to illustrate a sort of reverse mortality gradient for suicide. Suicide rates rise with classes, going from 13 among unskilled to 20.1 among lower managers. While opposite recent empirical findings, it aligns with the Durkheimian notion that poverty is protective against suicide risk. The effect diminishes as it reaches higher managers, who had a suicide rate of 14.3. Going by the rates, belonging to the elites seemed to offer some sort of protection, whether this was through being isolated from the disruptive aspects of society or a perceived high utility is hard to say.

Surprisingly, looking at the rates for different household size, shows that those who lived alone had the lowest suicide rate, apart from the institutional households. This would be contrary to Durkheimian theory of social integration, in that social isolation seems protective simply looking at the rates. However, having the option to live alone might be regarded a luxury at this time, given a very crowded city and thus could in itself reduce stress.

Suicide rates for residential mobility reveal a certain pattern. Those who never moved had a substantially lower suicide rate than those who did. More intense mobility seems to be associated with an increased suicide rate, up to a certain point. The effect is diminished when moving more frequently than 1 time a year, although the suicide rate is still high. There seem to be a disintegratory relationship up to a certain intensity, where perceived social dislocation accumulates with each additional move. If an individual is moving more than this, it is high likely that moving is part of the perceived norm thus no longer breaking previous attachments.[121] Either by actively living a mobile life or bringing with you the social group that you are a part of, thus intense mobility would have less of an effect on social integration. Acute social dislocation does not seem to have any particular effect in terms of suicide risk among those who were not born in Stockholm. For internal migrant the suicide rate was the same as for the Stockholm born. For foreign migrants, the suicide rates were even lower. Potentially, it could be indicative of the "healthy migrant" wherein larger migrant population exhibit better physical and mental health compared to native populations, despite adverse factors.[122]

6.2. Model analysis

In table 6, the results of the main Cox model, stratified on social class, are showcased. Models use age as a time scale, but age groups are furthermore included in the models to account for

[121] This could also indicate that one's identity is less tied to physical places. Abrutyn, "Toward a General Theory of Anomie", p. 126; Durkheim, Suicide, pp. 216-217.
[122] Chen et al., "Socio-economic studies on suicide", p. 287.

age composition. First of all, no greater level of significance was reported for the birthplace categories, except for unknown. Foreign born skilled workers did however display lower levels of significance, at a 1.6 HR, meaning that they faced an estimated 60 percent higher risk of suicide than internal migrants. Overall, estimates for birthplace indicate that depending on social class, a foreign migrant might be impacted differently by the acute social dislocation depending on what they leave behind.

HR's for residential mobility ranged between 5.6 for higher managers to 11.6 for lower managers. This meaning that lower managers who on average moved 1,7 times a year, had eleven times the risk of suicide compared to non-movers. Confidence interval between classes also seem to overlap, indicating that estimates are relatively close to one another. Although it should be noted that higher manager displayed quite a large interval. Residential mobility was significant at the same level across all classes, except for higher managers, which reported one level below. They also displayed the lowest HR out of all the classes. Lower managers and professionals displayed the highest HR in residential mobility. However, one should also consider that for the unskilled and skilled group, age was less capable of explaining variations in suicide risk. Given age, and increasing age's, association with suicide, this would mean that other significant variables should be regarded as even more important in explaining suicide risk. Either way, residential mobility exhibits signs of chronic dislocation, with a disintegrating effect being more potent with continuous moves.[123]

Weaker significance was found among the macro level variables, with higher managers and professionals displaying no significance for any of these. In-migration was the exception and showed higher levels of significance for lower skilled and lower managers, somewhat lower for the unskilled workers. Previous studies considering population measures as proxies for urbanization/modernization found a relationship with suicide, although considering suicide rates and furthermore, national rates, thus having accounted for rural areas.[124] Current results show that for a majority of the population, suicide risk increased by 0,01 percent each time in-migration rate rose by one unit.

[123] This result stands in contrast to Hagedoorn and Helbich's results which found that non-movers had more risk than movers. Hagedoorn & Helbich, "Longitudinal effects of physical and social neighbourhood change on suicide mortality", p. 7.

[124] Stack found that urbanization, although measuring it differently, was associated with a 0.19 percent increase in suicide in Finland, 1800-1985. Stack used proportion of urban population rather than population growth. Stack, "The Effect of Modernization on Suicide in Finland: 1800-1984.", pp. 145-146.

Table 6: Hazard Ratios for Suicide Risk by Social Class

Cox Proportional Hazards Models — Stockholm 1878-1925

	Unskilled workers N=1,073,779 Suicides=551	Lower skilled N=1,184,832 Suicides=692	Skilled workers N=814,199 Suicides=696	Lower managers N=804,980 Suicides=764	Higher managers N=224,192 Suicides=196
Covariates	HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)
Sex					
Male (ref: Female)	4.52 (3.73-5.49)***	4.22 (3.54-5.01)***	5.45 (4.27-6.94)***	6.58 (5.31-8.15)***	4.03 (2.73-5.95)***
Birthplace (ref: Internal migrant)					
Stockholm born	0.98 (0.79-1.2)	0.97 (0.81-1.15)	1.1 (0.92-1.31)	1.14 (0.97-1.33)	0.99 (0.71-1.38)
Foreign born	1.17 (0.52-2.65)	0.67 (0.32-1.42)	1.6 (1.07-2.4)*	0.71 (0.46-1.1)	0.79 (0.37-1.71)
Unknown birthplace	3.09 (1.25-7.67)*	4.03 (1.65-9.85)**	3.55 (1.74-7.26)***	1.54 (0.63-3.78)	2.52 (0.78-8.12)
Household size (ref: Alone)					
Small (1-4)	1.15 (0.93-1.41)	0.88 (0.74-1.05)	0.79 (0.67-0.94)**	0.75 (0.64-0.88)***	0.7 (0.5-0.97)*
Medium (5-9)	1.19 (0.88-1.62)	0.87 (0.66-1.16)	0.85 (0.66-1.1)	0.68 (0.53-0.89)**	0.77 (0.47-1.28)
Large/Institutional (10+)	1.4 (0.82-2.39)	0.48 (0.22-1.08)†	0.65 (0.35-1.19)	0.63 (0.4-0.98)*	0.67 (0.33-1.38)
Residential mobility					
Log mobility rate	9.9 (6.52-15.04)***	10.22 (6.74-15.49)***	8.55 (5.62-13.02)***	11.63 (7.56-17.89)***	5.61 (1.59-19.79)**
Age group (ref: 14-29)					
Age 30-49	1.48 (0.98-2.24)†	1.79 (1.26-2.56)**	1.4 (0.99-1.99)†	1.82 (1.28-2.59)***	1.92 (0.96-3.87)†
Age 50+	1.39 (0.8-2.42)	2.11 (1.22-3.63)**	1.84 (1.15-2.94)*	2.71 (1.69-4.34)***	2.13 (0.84-5.43)
Macro-level controls					
In-migration rate	1.01 (1-1.02)**	1.01 (1-1.01)***	1 (1-1.01)†	1.01 (1-1.01)***	1.01 (1-1.01)
Nuptiality rate	0.85 (0.75-0.97)*	0.88 (0.78-0.99)*	0.9 (0.8-1.01)†	0.88 (0.79-0.99)*	0.98 (0.77-1.25)
Wage cycle (HP)	1.02 (1-1.04)†	1 (0.99-1.02)	1.01 (0.99-1.02)	1 (0.98-1.01)	1.01 (1-1.03)
GDP growth	1.04 (1-1.07)*	1 (0.98-1.03)	1 (0.97-1.03)	1.01 (0.99-1.04)	0.99 (0.95-1.03)

*** p<0.001 ** p<0.01 * p<0.05 † p<0.10

Residential mobility = log(1 + moves per year of observation). HR represents hazard ratio per unit increase in log mobility rate, with zero moves per year as implicit baseline.

Sex included as covariate (ref: Female). All models use age as time scale with age group as covariate. Robust standard errors clustered on individual.

Wage cycle = HP-filtered cyclical component of real wage index (λ=100). GDP growth = annual real GDP volume growth rate, Sweden.

Individual level HR should be interpreted according to reference group. An HR of 2.0, indicates that the group had double the risk of suicide compared to reference group.

Regarding the aggregated economic variables, they only showed significance for the unskilled group. This would imply that only lowest classes felt the anomic presence of economic fluctuations. Interestingly enough, both variables displayed a HR value above 1 for both covariates for this group. Meaning that there is an indication of a pro-cyclical relationship with anomie. Years of positive economic growth would have produced a positive or neutral effect amongst the other classes in terms of mental wellbeing. For the unskilled workers, economic growth might have entailed a sudden change in industrial activity and their working environment. Although for other classes, this might have entailed new opportunities, unskilled workers might have had less means to reap the positive benefits. Another interpretation could be that sudden economic upswings might have lured with the promise of a better life, setting the unskilled workers up for disappointment.[125] This, indicating that they are less equipped to deal with the lures of prosperity. The urban centres of the time promised economic welfare compared to rural communities, and many were drawn to Stockholm for just this reason.[126] Having this promise broken may very well have been a cause for anomie.

Nuptiality control was found to be significant on the lower end across the board. While it should be cautiously interpreted due to the aforementioned issues discussed in the Data chapter, an increase in marriages seem to protect against suicide risk, indicating stronger integration. With each additional marriage entered per 1000 of population, suicide risk decreased between 10 to 15 percent.

Household size controls showed no significance among lower classes. For skilled workers and above, a small household of 1-4 additional members seem to protect somewhat against suicide risk. This effect was more pronounced among lower managers and professionals, where significance levels for a small household were at $p<0.001$, levels which declined for larger households than that. To be noted is that the largest household category, Large/Institutional, likely entails different factors for lower managers than it does for unskilled workers. Where for unskilled workers, the institutional category might represent a lot of poorhouses, for lower managers it most likely indicates simply large families. Either way, as per Durkheim, the results indicate that social isolation reflects a lessening of social

[125] While no sweeping inference should be made regarding individual cases, strain theory as per Merton and social disjunction as per Abrutyn claim that unequal distribution of means which to achieve aspirations is cause for anomie. Abrutyn, "Toward a General Theory of Anomie", pp. 125-126; Merton, "Social Structure and Anomie.", pp. 681-682.

[126] According to the push and pull framework, unemployment and prices would push people out of the area of origins. Pull factors at destination would job opportunities and wages. Hedefalk, Svensson & Tegunimataka "Migration in Times of Economic Growth and Recession", pp. 116-117.

integration.[127] Interestingly enough, while non-significant and should be taken with a grain of salt, unskilled workers consistently displayed a HR above 1, for households consisting of more than one person. As with the institutional households, a single household could very well be two different things between those of lower classes and higher. For an unskilled worker, a single household might very well reflect more economic stability and a getaway from the overcrowded conditions where people of the same class made their residence, thus offering a fixed physical space counteracting the effects of social dislocation. For a lower manager, the single household is rather indicative of social isolation and mayhap a loss of status.[128]

6.3. Robustness and sensitivity

This chapter presents the results of the robustness testing. In table 7, results from the Cox models are compared to results from the Fine & Gray models, which accounts for competing risks. Results of the F&G model highlighted in grey. Age groups were included in the models but removed in the table due to results being incomparable between the two models. Because the F&G models keep other causes of death in the risk set, the age covariate becomes inflated.

When accounting for competing risks, most findings seem robust, meaning HR and SHR's are virtually similar and significance level are the same. For birthplace, the inflated results of the unknown covariate seem to lessen its effects. Macro level controls are seemingly also robust to competing risk. Nuptiality increased in significance for unskilled and skilled workers. Household size saw some significant differences. The protective effect of household consisting of more than one person for the higher classes was attenuated, although still displays hazards of less than 1. On the other hand, for unskilled workers, larger households now show significance, strengthening the finding that different household sizes entail separate things across the classes.

Most striking may be the decreasing hazard ratios of the residential mobility variable. SHR's are substantially lower than the HR's for each social class, almost decreasing by as much as 80 percent for the lower skilled group. Where moving 1.7 times a year was estimated to result in an eleven times higher risk, the same moving rate in the F&G model is estimated to results in slightly less than three times the risk. Results are, however, still significant even after

[127] Durkheim, *Suicide*, p. 169.

[128] Living in a household by oneself does not necessarily have to indicate social isolation. Isolation could furthermore be considered a consequence of social integration, rather than a cause. Furthermore, it has been suggested that men benfitted from holding positions of authority outside and within the home. Living alone would deprive a man of this authority, either a further consequence of total disintegration or an active process. Abrutyn, "Toward a General Theory of Anomie", p. 114; Tilly, "Women, Women's History, and the Industrial Revolution.", pp. 131–132

competing risk, except for higher managers. Distribution of hazard furthermore now tells a different story. Unskilled workers are seemingly most vulnerable to the effects of intense mobility during the period, when accounting for competing risk. For the middle classes, a reverse gradient of sorts becomes clear, with the lowest risk among lower skilled to highest among lower managers.

Table 7: Hazard and Subdistribution Hazard Ratios for Suicide Risk by Social Class

Cox Proportional Hazards and Fine & Gray Models — Stockholm 1878-1925

	Unskilled workers N=1,073,779 Suicides=551		Lower skilled N=1,184,832 Suicides=692		Skilled workers N=814,199 Suicides=696		Lower managers N=804,980 Suicides=764		Higher managers N=224,192 Suicides=196	
Covariates	Cox HR	F&G SHR	Cox HR	F&G SHR	Cox HR	F&G SHR	Cox HR	F&G SHR	Cox HR	F&G SHR
Sex										
Male (ref: Female)	4.52 (3.73-5.49)***	4.241 (3.501-5.138)***	4.22 (3.54-5.01)***	3.507 (2.947-4.175)***	5.45 (4.27-6.94)***	5.929 (4.636-7.583)***	6.58 (5.31-8.15)***	5.593 (4.513-6.931)***	4.03 (2.73-5.95)***	4.167 (2.813-6.173)***
Birthplace (ref: Internal migrant)										
Stockholm born	0.98 (0.79-1.2)	1.082 (0.885-1.322)	0.97 (0.81-1.15)	0.998 (0.84-1.186)	1.1 (0.92-1.31)	1.166 (0.981-1.387)†	1.14 (0.97-1.33)	1.119 (0.958-1.308)	0.99 (0.71-1.38)	1.055 (0.761-1.463)
Foreign born	1.17 (0.52-2.65)	1.223 (0.542-2.757)	0.67 (0.32-1.42)	0.715 (0.339-1.51)	1.6 (1.07-2.4)*	1.602 (1.069-2.401)*	0.71 (0.46-1.1)	0.735 (0.478-1.129)	0.79 (0.37-1.71)	0.811 (0.376-1.751)
Unknown birthplace	3.09 (1.25-7.67)*	2.731 (1.103-6.762)*	4.03 (1.65-9.85)**	2.14 (0.88-5.205)†	3.55 (1.74-7.26)***	2.639 (1.292-5.39)**	1.54 (0.63-3.78)	0.753 (0.307-1.844)	2.52 (0.78-8.12)	1.696 (0.528-5.443)
Household size (ref: Alone)										
Small (1-4)	1.15 (0.93-1.41)	1.194 (0.98-1.455)†	0.88 (0.74-1.05)	0.987 (0.834-1.169)	0.79 (0.67-0.94)**	0.881 (0.743-1.043)	0.75 (0.64-0.88)***	0.861 (0.732-1.012)†	0.7 (0.5-0.97)*	0.729 (0.519-1.024)†
Medium (5-9)	1.19 (0.88-1.62)	1.331 (0.98-1.808)†	0.87 (0.66-1.16)	1.035 (0.782-1.371)	0.85 (0.66-1.1)	1.011 (0.781-1.309)	0.68 (0.53-0.89)**	0.824 (0.634-1.072)	0.77 (0.47-1.28)	0.86 (0.513-1.444)
Large/Institutional (10+)	1.4 (0.82-2.39)	1.763 (1.033-3.009)*	0.48 (0.22-1.08)†	0.632 (0.281-1.418)	0.65 (0.35-1.19)	0.834 (0.452-1.536)	0.63 (0.4-0.98)*	0.893 (0.57-1.401)	0.67 (0.33-1.38)	0.893 (0.432-1.844)
Residential mobility										
Log mobility rate	9.9 (6.52-15.04)***	3.104 (1.998-4.822)***	10.22 (6.74-15.49)***	2.259 (1.476-3.457)***	8.55 (5.62-13.02)***	2.498 (1.611-3.874)***	11.63 (7.56-17.89)***	2.746 (1.801-4.187)***	5.61 (1.59-19.79)**	1.665 (0.483-5.748)
Macro-level controls										

	Unskilled workers N=1,073,779 Suicides=551		Lower skilled N=1,184,832 Suicides=692		Skilled workers N=814,199 Suicides=696		Lower managers N=804,980 Suicides=764		Higher managers N=224,192 Suicides=196	
Covariates	Cox HR	F&G SHR	Cox HR	F&G SHR	Cox HR	F&G SHR	Cox HR	F&G SHR	Cox HR	F&G SHR
In-migration rate	1.01 (1-1.02)**	1.009 (1.003-1.015)**	1.01 (1-1.01)***	1.008 (1.004-1.013)***	1 (1-1.01)†	1.005 (1-1.009)*	1.01 (1-1.01)***	1.007 (1.003-1.011)**	1.01 (1-1.01)	1.006 (0.998-1.013)
Nuptiality rate	0.85 (0.75-0.97)*	0.806 (0.71-0.915)**	0.88 (0.78-0.99)*	0.865 (0.767-0.975)*	0.9 (0.8-1.01)†	0.854 (0.761-0.957)**	0.88 (0.79-0.99)*	0.87 (0.775-0.977)*	0.98 (0.77-1.25)	0.935 (0.735-1.189)
Wage cycle (HP)	1.02 (1-1.04)†	1.018 (0.997-1.039)†	1 (0.99-1.02)	1.002 (0.988-1.016)	1.01 (0.99-1.02)	1.007 (0.99-1.024)	1 (0.98-1.01)	0.999 (0.984-1.013)	1.01 (1-1.03)	1.014 (0.993-1.035)
GDP growth	1.04 (1-1.07)*	1.039 (1.005-1.074)*	1 (0.98-1.03)	1 (0.974-1.026)	1 (0.97-1.03)	0.999 (0.97-1.029)	1.01 (0.99-1.04)	1.012 (0.986-1.039)	0.99 (0.95-1.03)	0.986 (0.944-1.031)

*** p<0.001 ** p<0.01 * p<0.05 † p<0.10

Residential mobility = log(1 + moves per year of observation). HR/SHR per unit increase in log mobility rate (implicit baseline: zero moves/year).

Age group included as covariate in all models (not reported). F&G age coefficients are not comparable to Cox estimates due to differences in the subdistribution hazard framework.

F&G models for Lower skilled and Lower managers used 20% random sample of censored observations to achieve numerical convergence.

Sex as covariate (ref: Female). Robust SEs clustered on individual in Cox models. Wage cycle = HP-filtered real wage cyclical component (λ=100). GDP growth = annual real GDP volume growth, Sweden.

Finally, sensitivity checks were made with different stratifications than on social class (see appendix B.1 and B.2). Estimations were produced through Cox models with one being stratified on age group and one being stratified on age group *and* sex. From these, it is possible to discern that social class in and of itself was not able to explain variations in suicide risk for the most part. Although, when stratifying on sex and age, lower managers had an inverse significant relationship across sex, with male lower managers experienced 50 percent higher risk and female lower managers 50 percent lower risk, compared to the reference group. Generally, younger people, especially men, were seemingly at more risk to the effects of social disruption. They displayed the highest HR towards residential mobility. Although, the middle-aged group displayed quite a high HR as well, the effect declined for the oldest group. However, all groups were significant. The youngest group were also most vulnerable to economic fluctuations where for younger men specifically, the wage cycle displayed significance at p<0.001. The association was positive, as it was for unskilled workers, indicating that they were vulnerable to the socially disruptive effect of times of prosperity. This

indicate that as economic conditions and living standards improve, the risk for suicide increased among young people.

6.4. Discussion

The analysis set out to find which factors impacted suicide risk during Stockholm's industrialization and how said risk was distributed across classes. The variables chosen were guided by a foundation in Durkheimian theory on suicide along with Abrutyn's reformulation of anomic theory, through the concepts of social disruption and social dislocation. The results of the descriptive analysis found a reverse gradient in terms of social class, with the suicide rate being lowest among unskilled workers and highest among lower managers and professionals. According to Durkheim, those impoverished would find themselves protected against the effects of anomie. An individual's desires are controlled by their resources, thus the aspirations of those in poverty are tempered.[129] However, as previous research on suicide and poverty, this relationship is usually confounded by aspects such as alcoholism, unemployment and physical health.[130] The reverse gradient in suicide rate could suggest that lifestyle behaviour associated with adverse suicide risk was more common among higher classes. Especially as lifestyle behaviour was found to be the main explanation for why the SES gradient emerged as late as it did.[131]

Increased rates of residential mobility displayed increasing suicide rates up to a certain point. The rate peaked for those who moved between 1 time a year and 1 time each two years at 33.7. The individuals who moved more than once a year had a somewhat lower rate, but it should still be regarded as high compared to the non-movers, at 21.5. Understanding this through the concept of dislocation entails speculating about what sort of individuals these were. The breaking point could indicate that the category of very frequent movers, contains those who were used to living a mobile life and that their social identity was independent of residence and not being tied to a physical place.[132] Either way, those who never moved had substantially lower rates than those who did, indicating that high residential mobility should be regarded as disrupting of social life in some way.

[129] Durkheim, Suicide, p. 214.
[130] Stack, "Suicide: A 15-Year Review of the Sociological Literature Part I", pp. 154-155.
[131] Debiasi & Dribe "SES inequalities in cause-specific adult mortality", pp. 1052-1053.
[132] Abrutyn suggest that social dislocation does not necessarily have to point to negative consequences. He exemplifies through cases where dislocation caused dislocated groups to channel efforts into establishing new junctions of solidarity. The examples he lists are however more indicative of acute, rather than chronic anomie. Abrutyn, "Toward a General Theory of Anomie", pp. 126-127.

Apart from the in-migration rate, which as mentioned previously should be treated with care, the Cox models showed that residential mobility was the variable best suited in explaining suicide risk. Estimates were similar across classes, excepting higher managers who although had an overlapping confidence interval. The Cox model showed a dramatic increase in between eight- and eleven-times higher risk across classes, per unit increase in log-transformed moves per year, which as previously mentioned translates into 1.7 moves a year.

Of concern is the potential that residential mobility may be endogenous to mental health deterioration. If an individual is experiencing psychological stress, then they may be more likely to move more frequently, thus rendering mobility a symptom rather than a cause. To be noted however, is that several features of the results seem to mitigate concerns of these issues. Regarding issues of reverse causality, the variable is measuring suicide risk in regard to cumulative moves, not just recent moves. If the results are indicating reverse causality, then this means some individuals are existing in pre-suicidal states for more than ten years. Furthermore, the gradient results are pointing to something else. It would be reasonable to expect a more abrupt threshold in the case of reverse causality. Looking at the sensitivity check for age groups (see appendix B.1), the effect is more concentrated among young people results which a pure reverse causal relationship would not produce.

The main issue lies in omitted variable bias. Social class controls for occupational position, which may reflect occupations more prone to volatile employment status. Stressful aggregate economic conditions are also accounted for by the wage cycle and GDP. The study cannot however control for individual-level employment status and its confounders, such as alcoholism and physical illness.[133] For these reasons, the findings on mobility cannot report a true causal relationship. It could however reflect broader social consequences of residential instability.

The estimates for the birthplace covariates showed no substantial significance level, suggesting that acute social dislocation was not as apt in explaining suicide risk as chronic dislocation. Integratory factors in terms of household size showed some interesting results. For the higher classes, there seemed to be a protective effect in living in a household consisting of more people, especially for lower managers. Even if somewhat weak, the results would suggest that household size had an impact on the effects social isolation has on integration. This result was found to not be robust when accounting for competing risk. Although also of weaker significance, the robust test found that unskilled workers were at more risk if they were living

[133] Stack, "Suicide: A 15-Year Review of the Sociological Literature Part I", pp. 153-155.

in more crowded households. From the results, it is hard to say if the crowded conditions of larger households were risk inducing or if the single households signal a stability not commonly found among the unskilled workers. Either way, results could suggest that social integration operates under different circumstances dependent on social class.

The disruptive effects of a moving economy were mainly felt by the lowest class, though to a limited extent. When testing for sensitivity, it was found that younger men were most particularly vulnerable to the wage cycle. The main conclusion to be drawn from the macro level factors overall, is that individual level indicators were better able to explain suicide risk for most groups.

In turn, only minor support was found for the perceived utility function. Age group composition displayed higher HR's and significance for most groups, it did not do so for the unskilled workers. If social class to an extent can roughly represent income and if utility is a function of lifetime income and age, then unskilled workers might perceive their utility to be at 0 at a much earlier age.[134] Of course, acknowledging that income has here not been measured. Support for the economic variables was found among the unskilled workers, however both significance and estimates were low. When on the other hand testing for sensitivity toward age, the wage cycle variable displayed high levels of significance.

To conclude: chronic social dislocation was found through intense residential mobility to best explain suicide risk in the models and displayed a clear gradient in suicide rates. However, there is likely more to the association than what the results can show. In interpreting these results, one should be aware of both reverse causality and omitted variable bias. However, the strong association, which held against both robustness and sensitivity tests, do indicate that it is an association worth exploring. The fact that unskilled workers were more, albeit moderately, vulnerable overall and to macro level effects but at the same time had the lowest suicide rate, suggest that the results are not displaying the full picture.

[134] HISCLASS should only be interpreted as rough estimates of general income divisions and is no true measure of income. van Leeuwen & Maas, *HISCLASS: A Historical International Social Class Scheme*, pp. 26-27.

7. Conclusions

The purpose of this study was to contribute to the study of inequality between social classes during periods of industrialization in terms of mental wellbeing. Economic history has a long track-record of studying inequality in terms of living standards, income, wealth and mortality but has dedicated less attention to mental health. But not without good reason, since data on mental health is hard to come by. In order to be able to investigate this, the study used suicide risk as a proxy. Thus, the study sought the answer the question of what impacted suicide risk in Stockholm during the period 1878-1925 and who were at risk. In taking advantage of Stockholm's rich archival resources, it was possible to procure highly detailed individual level data on those who had committed suicide and their contemporaries. To determine risk, a Cox proportional hazards model was employed. Explaining factors investigated included aggregated economic change, population growth through in-migration and mobility, both migratory and residential. These research design decisions were guided by classical sociological theory on social integration and disruption.

The results of the Cox model showed that residential mobility was the most impactful driver of suicide risk. This result held between classes and robustness testing. Complementing the Cox model, a descriptive analysis was also made, where it was shown that increasing residential mobility resulted in higher suicide rates up a certain limit. In-migration rate was also found to be a significant explainer for most social classes. However, since the in-migration rate was included as a proxy of urbanization due to classic Durkheimian theory, care should be taken when interpreting these results. Previous empirical findings by Anderson indicate that industrialization, and in turn urbanization, are not reliable indicators of what drove suicide risk during this period. [135] In ascribing too much value to this finding, one runs the risk of fortifying notions of modern society being associated with moral degeneration and deviance.

As for differences between classes, when accounting for competing risk, unskilled workers were most vulnerable to chronic social disruption of residential mobility. Although, confidence intervals largely overlapped among the classes, indicating that real risk estimates likely are similar. However, they were the only social class which displayed any sensitivity toward the aggregated economic variables. Furthermore, they showed a positive association with these, meaning that when GDP and the wage cycle were booming, suicide risk among the

[135] Since there is no rural data to compare to, it is not possible to say if it was the cities or rural areas that drove the suicide rates nationally. However, the argument that industrialization did not impact suicidal behaviour seem to not be applicable in the case of Stockholm. Anderson, "Did Suicide Increase with Industrialization in Victorian England?", pp. 165-166.

unskilled workers increased. In their study on infant mortality in Stockholm which also used data from Rotemannen, Molitoris and Dribe concluded that inequality persisted while living standards increases for all. Workers seemed to be most vulnerable to economic stress in terms of child- and infant mortality, reflecting the vulnerability found for workers in present study.[136] However, the positive association that unskilled worker's suicide risk had with the wage cycle and GDP suggest other mechanisms. Whereas improved living conditions might have been positive in terms of child- and infant mortality, the results imply that the transition into better conditions is stressful in and of itself.

Despite that the unskilled workers seemed most vulnerable to aggregated effects, the other classes displayed higher suicide rates overall, producing a gradient effect from unskilled workers up and until lower managers. The most glaring difference in risk between classes was found for the highest class. After robustness testing, higher managers only displayed very slight significance for the protective effects of living in a small household compared to living alone. Taken alongside the results of the descriptive analysis, where they had the second lowest rate, higher managers seem protected against socially disruptive effects.

In controlling for household size, the analysis found that for classes above unskilled, living by yourself in a household was associated with higher suicide risk. For unskilled workers, the relationship was inverse, with living in a single household being comparatively protective against suicide risk. In its entirety, whereas the results showed similar risk estimates for the majority of classes, there are still questions left unanswered. The finding on household indicates that there could be fundamental different mechanisms driving suicide risk between classes. This suggests that the study of material wellbeing in investigating factors such as living standards, wages and mortality, do not tell the full story of wellbeing during the industrialization. Mental wellbeing, in this case illustrated as suicide risk, was to an extent distributed differently and for some, industrialization may have been the culprit.

While the difference in risk between most classes could be considered marginal, the study makes two primary contributions of interest; one, residential mobility was a consistent and robust predictor of suicide risk and, two, the highest class seems protected against the detrimental effects of social disintegration. The thesis contributes to the economic historian tradition of studying inequality and industrialization by showing that mental wellbeing can be

[136] While short-term fluctuations in FAW, seemed to indicate no discrimination in child mortality between classes, the relative difference between classes remained. They reason that the working class was most sensitive to common infections, which then spread to other classes. Molitoris & Dribe "Industrialization and inequality revisited", pp. 192-193.

differentiated by social class. Whereas it was not able to find the exact mechanisms that caused this, the study pointed to lifestyle behaviour potentially being the differing factor. It furthermore actively contributes to sociological study of suicide. Studies on residential mobility seldom consider the effect continuous moving, only focusing on the binary distinction of movers and non-movers. While historical context of the present study may influence the results, they suggest that there could be a very real underlying effect behind intense movement. Both of these contributions also indicate in which direction, future research should look.

Contrary to Hagedoorn and Helbich finding that adult movers experienced less suicide risk, the results of this study showed that moving was one of the main driving forces behind suicide risk during this period and place. They explained their results as due to movers having the agency to escape negative life circumstances and undesired neighbourhoods.[137] This should be contrasted to industrializing Stockholm, where a mover might have little or no agency at all in the decision. This is strengthened by the finding that higher managers, who most likely would have had said agency, were at no significant risk after robustness tests. The phenomena should however merit further investigation due to the aforementioned endogeneity issues. If speculating, one could assume it might be capturing periods of unemployment, or an aspect which has not been explored in this thesis, specific moving patterns. In the latter's case, a continued study could investigate factors such as segregation or ecological circumstances, such as access to green areas.[138]

Even though similar risk estimates was found for residential mobility between classes, the suicide rate gradient by class, showed that it is not simply a relationship of higher income or class equalling a better mental wellbeing, at least up to a point.[139] As has previously been mentioned, suicide is complex and relates to many factors and controlling for all is most likely impossible. Studies on the SES gradient showed that the most important differential between classes in determining the emergence of the gradient, was lifestyle behaviour and stress. The lifestyle choices such as alcoholism, substance abuse and inactivity, are in contemporary

[137] Hagedoorn & Helbich, "Longitudinal effects of physical and social neighbourhood change on suicide mortality", p. 6.
[138] Hagedoorn & Helbich, "Longitudinal effects of physical and social neighbourhood change on suicide mortality", pp. 5-6.
[139] Comparatively, if general mortality is indicative of other forms of wellbeing, as shown by Brea-Martinez, Hedefalk and Nilsson, simply living in wealthier areas could produce better health effects. Gabriel Brea-Martinez, Finn Hedefalk & Therese Nilsson, "Income, Inequality, and Geography" in *Urban Lives: An Industrial City and Its People During the Twentieth Century*, Martin Dribe, Therese Nilsson & Anna Tegunimataka (eds), (Oxford Academic, 2024), pp. 324-325. https://doi.org/10.1093/oso/9780197761090.001.0001

studies shown to be relevant predictors of suicide.[140] If lifestyle choices had so much of an impact as to suppress the mortality gradient up until recently, it makes a case for investigating these in relation to suicide risk. Future studies on historical suicides should consider investigating differentials in lifestyle behaviour, if possible, to see if it as well can explain the higher rates among middle-classes and the immunity towards socially disruptive phenomena among elites.

While the theoretical framework covered social disintegration through economic, demographic and social indicators, it did not control for socioemotional factors. Contemporary sociology highlights socioemotional impact on suicide risk and aspects such as personal loss in separation or bereavement were not investigated.[141] If it is possible to consistently establish relationships within the Roteman data then social disjunction could eventually be controlled for as well.

The biggest caveat in trying to generalize results of a study concerning suicide lies in the disparities in gender. As models were stratified on sex, the effect is assumed to be similar between men and women. The conclusion one can draw from the results is that women were of course not immune to the socially disorienting effects of industrialization, as postulated by Durkheim and his contemporaries.[142] To fairly assess a similar situation for women, other measures than suicide would be more appropriate, be it suicide attempts or prevalence of depression. Measures which of course are much harder to come by in studying this period.

While the data used for the analysis has to be regarded as a sample, and not the full population of suicides in Stockholm, quality checks indicated that the sample was well-suited. In any case, having the full population found in the statistical yearbooks would still most likely not cover all of the suicides in a given time and place, especially in a historical context. It is a known issue in studies looking at historical suicide that they will inevitably encounter problems of under registration.[143] Although there are overall implications for this, the data issues faced in this study is not unique and ultimately, should not hurt comparability to other research. It also has to be underscored that the conditions of Stockholm observable in the data are not translatable into circumstances today. Of course, social isolation and economic fluctuations

[140] Alcohol consumption and substance abuse for example being such. While hard to apply in historical studies, other cultural and behavioural factors identified include imitation and subcultural characteristics. Stack, "Suicide: A 15-Year Review of the Sociological Literature Part I", pp. 153-155.

[141] While the current study has used part of Abrutyn's framework, an important feature of it is the operationalization of shame as a socioemotional aspect of anomie. Abrutyn, "Toward a General Theory of Anomie", pp. 126-127

[142] Kushner, "Suicide, Gender and Fear of Modernity", p. 467-469; Durkheim, Suicide, p. 263.

[143] Pescosolido, "Bringing Durkheim into the Twenty-First Century", pp. 268-269; Axelsson, *Självmord i teorin*, p. 20.

continue to show associations with suicide, the intense migration and mobility could be more unique for processes of industrialization.

To summarize, this thesis contributes to the study of industrialization by showing that the processes which lead to the development of material wellbeing, had a negative impact on select group's mental wellbeing in terms of suicide risk. The results indicate that while living conditions might have overall improved for those of less means, the journey there is potentially distressful. The fact that people of less means experience greater hardship than those of more, is not necessarily a new finding, but those of more means were shown to be just as vulnerable, if not more, to socially disruptive effects. Furthermore, it was not simply due to a moral degeneration, but concrete consequences born out of the transforming city. So, when economic historians are asking the question of how wellbeing evolved during the industrializations, it is not only a matter of improved living conditions or less infant mortality. It should also be a matter of the very human experiences that lead there.

8. Declaration on the use of AI

Claude Sonnet 4.6 was used in constructing code in RStudio for the analysis as well as visualization. Claude was also used for proofreading.

9. Bibliography

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9.2. Literature

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Appendices

Appendix A: Annual data

Wage data taken from:

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Table 8. (A.I.) Annual Macro-Level Variables, Stockholm 1878-1925

Raw annual data for all macro-level controls used in the Cox proportional hazards models

	Stockholm municipality			Sweden (national)		
Year	Population	Marriages (n)	Nuptiality rate	Real wage index	CPI	GDP growth (%)
1878	161,722	1,482	9.2	53.6	81.7	2.11
1879	163,040	1,502	9.2	52.4	77.9	0.20
1880	167,868	1,481	8.8	52.1	82.6	2.04
1881	174,652	1,508	8.6	53.2	84.3	-0.06
1882	182,358	1,624	8.9	55.5	83.7	3.06
1883	190,842	1,816	9.5	56.7	82.0	3.49
1884	200,781	1,912	9.5	59.7	79.0	2.22
1885	211,138	2,071	9.8	62.7	74.9	2.04
1886	216,807	1,857	8.6	64.1	71.0	-0.61

	Stockholm municipality			Sweden (national)		
Year	Population	Marriages (n)	Nuptiality rate	Real wage index	CPI	GDP growth (%)
1887	221,551	1,984	9.0	68.4	66.9	-0.71
1888	228,118	1,959	8.6	66.1	71.0	2.42
1889	236,350	1,959	8.3	67.9	73.2	3.00
1890	245,331	2,046	8.3	68.3	74.4	3.09
1891	248,051	1,982	8.0	65.2	79.3	2.68
1892	249,252	1,931	7.7	69.0	76.0	1.13
1893	252,937	1,875	7.4	72.8	73.9	0.64
1894	259,304	1,871	7.2	77.4	70.0	0.84
1895	267,100	1,899	7.1	78.3	70.1	6.15
1896	274,609	1,995	7.3	80.6	70.6	5.65
1897	282,795	2,213	7.8	80.8	73.8	3.20
1898	291,589	2,415	8.3	82.9	76.3	3.61
1899	297,078	2,535	8.5	81.6	80.8	4.10
1900	301,322	2,610	8.7	83.0	83.0	1.36
1901	302,397	2,392	7.9	83.5	82.6	-1.11
1902	305,706	2,375	7.8	84.7	82.9	0.05
1903	310,147	2,411	7.8	85.9	82.6	5.01
1904	318,446	2,600	8.2	88.9	82.6	3.08
1905	324,988	2,810	8.6	90.2	84.0	0.96
1906	333,238	3,080	9.2	91.8	86.9	8.05
1907	337,960	3,217	9.5	92.4	91.3	4.11
1908	340,082	3,031	8.9	92.7	93.2	0.77
1909	342,316	3,066	9.0	96.6	91.8	-1.12
1910	344,332	3,223	9.4	101.1	91.7	7.76
1911	347,099	3,195	9.2	101.4	92.8	1.56
1912	350,955	3,233	9.2	100.0	97.7	4.64
1913	382,085	3,327	8.7	100.7	98.8	6.99
1914	386,270	3,347	8.7	103.8	100.1	-0.05
1915	392,427	3,434	8.8	94.6	115.0	1.84
1916	408,792	3,623	8.9	92.0	130.0	4.61
1917	413,163	3,659	8.9	92.5	164.0	-7.07
1918	408,456	3,867	9.5	90.4	241.0	-6.12
1919	415,201	3,797	9.1	108.8	266.0	4.87
1920	419,788	3,988	9.5	133.7	271.0	6.78
1921	422,042	3,949	9.4	162.0	221.0	-8.15
1922	424,944	3,787	8.9	140.7	184.0	7.81
1923	429,812	4,042	9.4	142.6	174.0	6.44
1924	438,896	4,017	9.2	145.1	174.0	4.05
1925	442,528	4,001	9.0	146.7	177.0	4.20

Population = registered municipal population of Stockholm. Used as denominator for nuptiality rate and in-migration calculation.

Nuptiality rate = marriages per 1,000 municipal population per year. Raw marriage count (nuptiality) divided by pop_muni × 1,000.

Real wage index = weighted average nominal hourly earnings of male manufacturing workers (Prado 2010), deflated by CPI and indexed to 1912 = 100. Source: Sveriges Riksbank Historical Monetary Statistics.

	Stockholm municipality			Sweden (national)		
Year	Population	Marriages (n)	Nuptiality rate	Real wage index	CPI	GDP growth (%)

CPI = Consumer Price Index, deflation index, 1914 = 100. Source: Sveriges Riksbank Historical Monetary Statistics

GDP growth = annual real GDP growth rate (%), Sweden. (Edvinsson, 2014)

Appendix B: Sensitivity checks

Table 9. (B.I.) Hazard Ratios for Suicide Risk by Age Group

Sensitivity Models — Cox Proportional Hazards, Sex Stratified Within Age Groups. Stockholm 1878–1925

Covariates	Ages 14–29 N = 1,928,588; Suicides = 723 HR (95% CI)	Ages 30–49 N = 1,328,899; Suicides = 1421 HR (95% CI)	Ages 50+ N = 489,041; Suicides = 749 HR (95% CI)
Birthplace (ref: Internal migrant)			
Stockholm born	1.17 (0.98–1.38)†	1.04 (0.93–1.17)	0.92 (0.78–1.1)
Foreign born	0.78 (0.44–1.36)	1.09 (0.78–1.52)	0.83 (0.51–1.35)
Unknown birthplace	5.75 (2.37–13.95)***	2.83 (1.5–5.35)**	1.65 (0.89–3.08)
Household size (ref: Alone)			
Small (1–4)	0.8 (0.67–0.95)*	*Stratified*	*Stratified*
Medium (5–9)	1.09 (0.83–1.43)	*Stratified*	*Stratified*
Large/Institutional (10+)	1.21 (0.79–1.84)	*Stratified*	*Stratified*
Residential mobility			
Log mobility rate	12.11 (9.11–16.11)***	10 (7.11–14.07)***	5.07 (2.98–8.63)***
Social class (ref: Unskilled workers)			
Lower skilled workers	1.04 (0.84–1.29)	1.03 (0.87–1.23)	1.03 (0.79–1.35)
Skilled workers	0.81 (0.63–1.03)†	0.87 (0.73–1.03)	1.18 (0.93–1.5)
Lower managers and professionals	1.1 (0.88–1.38)	1.09 (0.92–1.29)	1.24 (0.97–1.58)†
Higher managers and professionals	0.95 (0.65–1.4)	0.94 (0.73–1.2)	0.83 (0.59–1.17)
Farmers and farmworkers	0.98 (0.64–1.48)	0.92 (0.68–1.24)	0.87 (0.55–1.36)
Macro-level controls			
In-migration rate	1.01 (1–1.01)**	1.01 (1–1.01)***	*Stratified*
Nuptiality rate	0.8 (0.72–0.89)***	0.92 (0.85–0.99)*	0.88 (0.78–0.99)*
Wage cycle (HP)	1.02 (1.01–1.03)**	1 (0.99–1.01)	1.01 (0.99–1.02)
GDP growth	1.02 (1–1.05)†	1.01 (1–1.03)	0.99 (0.97–1.01)

Residential mobility = log(1 + moves per year). Sex stratified within all age groups via strata(sex). 'Stratified' indicates that variable was used as a stratification factor rather than a covariate; no hazard ratio is estimated. Household size stratified in ages 30–49 and 50+. In-migration rate stratified in ages 50+. Birthplace and social class included as covariates in all models. Standard errors clustered on individual. Efron method for ties.

Wage cycle = HP-filtered real wage cyclical component (λ=100). GDP growth = annual real GDP volume growth, Sweden.

*** p<0.001 ** p<0.01 * p<0.05 † p<0.10

Table 10. (B.II.) Hazard Ratios for Suicide Risk by Age Group and Sex

Sensitivity Models — Cox Proportional Hazards, Men and Women Modelled Separately. Stockholm 1878–1925

	Men N = 881,097; Suicides = 514	Women N = 1,047,491; Suicides = 209	Men N = 603,854; Suicides = 1160	Women N = 725,045; Suicides = 261	Men N = 176,733; Suicides = 590	Women N = 312,308; Suicides = 159
	Ages 14–29		Ages 30–49		Ages 50+	
Covariates	HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)
Birthplace (ref: Internal migrant)						
Stockholm born	1.21 (0.99–1.48)†	1.11 (0.81–1.53)	1.07 (0.94–1.22)	0.95 (0.72–1.26)	0.86 (0.7–1.05)	1.15 (0.82–1.62)
Foreign born	0.62 (0.3–1.26)	1.24 (0.5–3.07)	1.09 (0.76–1.56)	1.1 (0.48–2.52)	0.79 (0.46–1.35)	1.02 (0.32–3.25)
Unknown birthplace	5.62 (2.08–15.22)***	5.97 (0.83–42.83)†	2.88 (1.47–5.64)**	2.66 (0.38–18.84)	1.6 (0.8–3.2)	2.04 (0.52–7.9)
Household size (ref: Alone)						
Small (1–4)	0.65 (0.52–0.8)***	1.23 (0.9–1.7)	*Stratified*	*Stratified*	*Stratified*	*Stratified*
Medium (5–9)	1.04 (0.75–1.45)	1.22 (0.76–1.97)	*Stratified*	*Stratified*	*Stratified*	*Stratified*
Large/Institutional (10+)	1.02 (0.58–1.82)	1.53 (0.8–2.91)	*Stratified*	*Stratified*	*Stratified*	*Stratified*
Residential mobility						
Log mobility rate	12.62 (9.01–17.66)***	10.24 (5.89–17.81)***	10.88 (7.38–16.05)***	7.44 (3.58–15.49)***	5.78 (3.13–10.66)***	3.62 (1.26–10.38)*
Social class (ref: Unskilled workers)						
Lower skilled workers	1.37 (1.01–1.86)*	0.88 (0.62–1.25)	1.05 (0.85–1.29)	0.96 (0.7–1.32)	0.77 (0.56–1.06)	1.94 (1.18–3.18)**
Skilled workers	1 (0.73–1.36)	0.8 (0.47–1.36)	0.88 (0.72–1.08)	0.81 (0.53–1.23)	1.03 (0.79–1.34)	1.59 (0.95–2.68)†
Lower managers and professionals	1.58 (1.17–2.14)**	0.51 (0.31–0.85)**	1.12 (0.91–1.37)	0.94 (0.65–1.36)	1.11 (0.85–1.46)	1.48 (0.87–2.54)
Higher managers and professionals	1.15 (0.71–1.85)	0.88 (0.44–1.74)	0.94 (0.71–1.24)	0.97 (0.53–1.77)	0.68 (0.46–1.01)†	1.45 (0.7–2.98)
Farmers and farmworkers	1.29 (0.82–2.03)	0 (0–0)***	0.92 (0.67–1.27)	0.87 (0.21–3.53)	0.69 (0.42–1.14)	1.82 (0.7–4.76)
Macro-level controls						
In-migration rate	1.01 (1–1.01)*	1.01 (1–1.02)†	1.01 (1–1.01)***	1 (0.99–1.01)	*Stratified*	*Stratified*
Nuptiality rate	0.9 (0.79–1.02)	0.61 (0.49–0.76)***	0.9 (0.82–0.98)*	1.01 (0.85–1.21)	0.83 (0.73–0.95)**	1.07 (0.82–1.39)
Wage cycle (HP)	1.03 (1.01–1.04)***	1 (0.97–1.02)	1 (0.99–1.02)	0.99 (0.98–1.01)	1.01 (0.99–1.02)	1.01 (0.98–1.03)
GDP growth	1.03 (1–1.06)*	1 (0.96–1.05)	1.02 (1–1.04)†	0.99 (0.96–1.03)	1.01 (0.98–1.03)	0.94 (0.9–0.99)*

Residential mobility = log(1 + moves per year). Men and women modelled separately within each age group. 'Stratified' indicates that variable was used as a stratification factor rather than a covariate; no hazard ratio is estimated. Household size stratified in ages 30–49 and 50+ (covariate in 14–29). In-migration rate stratified in ages 50+. Birthplace and social class included as covariates in all models. Standard errors clustered on individual. Efron method for ties.

Wage cycle = HP-filtered real wage cyclical component (λ=100). GDP growth = annual real GDP volume growth, Sweden.

No female suicides observed among farmers/farmworkers aged 14–29; coefficient undefined.

*** p<0.001 ** p<0.01 * p<0.05 † p<0.10